Everything In Moderation: The Currency-Maturity Composition of Sovereign Debt*

Preliminary Draft

Raymond He†
The University of Texas at Austin

This version: November 1, 2026

Click here for the latest version.

Abstract

Emerging-market governments choose both the currency and maturity of their debt. I document a new fact: domestic currency sovereign debt has systematically shorter maturity than foreign currency debt. I develop a dynamic sovereign-debt model with endogenous default and inflation in which the government chooses among short- and long-term bonds denominated in domestic and foreign currency. Domestic currency and long-term debt provide insurance against bad states but create inflation, default, and dilution incentives that lower their prices today. Foreign currency and short-term debt provide stronger discipline but less insurance. When currency and maturity are chosen jointly, these forces make the extreme portfolios unattractive in normal states, leading governments to rely primarily on short-term domestic currency and long-term foreign currency debt. Calibrated to Türkiye, the model generates a domestic-to-foreign maturity ratio close to that observed in the data and shifts issuance toward short-term foreign currency debt in adverse states.

JEL Classification: E31, E43, E52, F34, H63
Keywords: currency, default, inflation, maturity, sovereign debt

*I am deeply indebted to Saroj Bhattarai, Olivier Coibion, Andrés Drenik, Oliver Pfäuti, Zachary Stangebye, Rachel Wellhausen, and Adrien Wicht for their invaluable guidance and feedback. I am grateful to Fernando Broner, Guido Lorenzoni, and Sergio Schmukler for sharing their data. I also thank Daniel Hsu, Hayden Parsley, and participants at the 2024 and 2026 Society for Computational Economics and the UT Austin Macro Writing Seminar for their helpful comments. Jacob Lawhern provided excellent research assistance. All errors are my own.

†E-mail: raymond.he@austin.utexas.edu

I Introduction

How should governments borrow? For emerging market economies, two central features of the debt portfolio are its currency denomination and its maturity. These choices are typically studied separately, even though governments choose them jointly and the underlying trade-offs are closely related. Domestic currency debt allows the government to reduce its real repayment burden through inflation in bad states, while long-term debt provides insurance because its market value falls when default risk rises. At the same time, both forms of insurance create incentives that must be priced by lenders: domestic currency debt creates inflation risk, while long-term debt creates dilution risk from future borrowing. This interaction matters empirically. Using sovereign bond issuance data for 19 emerging market economies from 1990 to 2024, I document that domestic currency debt has a much shorter average maturity compared with foreign currency debt, a pattern that is remarkably stable across countries and over time. To explain this fact, I study currency and maturity choice jointly and develop a quantitative sovereign-debt model to explain why governments systematically combine shorter maturities with domestic currency debt and longer maturities with foreign currency debt.

The main idea is that the hedging and incentive properties of currency denomination and maturity reinforce each other when the two margins are chosen jointly. Debt that is denominated in domestic currency or has long maturity provides insurance against adverse income and exchange-rate realizations, but also creates incentives for future inflation, default, or debt dilution that lower the price at which it can be issued. Foreign currency and short-term debt impose greater discipline on the government, allowing it to raise more revenue today, but provide less insurance against bad future states. As a result, the two most specialized instruments, long-term domestic currency debt and short-term foreign currency debt, are attractive mainly in extreme states. During normal times, governments prefer the more balanced combinations of short-term domestic currency debt and long-term foreign currency debt. Calibrated to Türkiye, the model generates a domestic-to-foreign maturity gap close to that observed in the data and reproduces governments' tendency to shift issuance toward short-term foreign currency debt in adverse states.

I begin by documenting three facts about sovereign debt issuance. The first, and the main empirical motivation for the paper, is that domestic currency sovereign debt has systematically shorter maturity than foreign currency debt. My bond-level data contains 24,910 issuances across 19 emerging market economies. The distribution of domestic currency issuance is concentrated below five years, while foreign currency issuance is concentrated at substantially longer maturities. The difference also appears within countries: foreign currency debt has a longer amount-weighted average maturity in 18 of the 19 countries in the sample. Finally, the pattern is present in every year of the 1990–2024 sample. Thus, the maturity difference is not driven by a small number of countries or particular episodes.

The second and third facts concern how both portfolio margins vary with borrowing conditions. When sovereign spreads are high, governments issue a smaller share of their debt in domestic currency and shorten the maturity of their issuance. A one-percentage-point increase in

the sovereign spread is associated with a 0.3 percentage-point decline in the domestic currency share of issuance and with declines of 0.05 and 0.1 years in the maturity of domestic and foreign currency bonds, respectively. The maturity response is larger for foreign currency debt. These patterns extend existing findings that domestic currency borrowing is procyclical and that governments shorten the maturity of foreign currency debt during periods of high spreads. Taken together, the evidence suggests that currency and maturity choices respond jointly to the state of the economy.

To understand these patterns, I build a dynamic model of a small open economy that borrows from competitive foreign lenders. A benevolent government chooses whether to repay or default, its inflation rate, and its debt portfolio. The government can issue four bonds: short-term domestic currency debt, long-term domestic currency debt, short-term foreign currency debt, and long-term foreign currency debt. The government cannot commit to future repayment, monetary policy, or borrowing decisions. Bond prices therefore reflect how the current portfolio affects future incentives to default, inflate, and issue additional debt.

The currency denomination of debt generates a trade-off between insurance and discipline. Domestic currency debt provides insurance because the government can raise inflation in bad states and reduce the real value of its repayment obligations. This state contingency is especially valuable when income is low and marginal utility is high. Lenders understand this incentive, however, and require compensation for the inflation risk associated with domestic currency debt. Foreign currency debt removes this inflation margin and can therefore be issued on more favorable terms. Its drawback is that the real repayment burden rises when the domestic currency depreciates, which in emerging markets tends to occur precisely when income is low. Domestic currency debt therefore provides greater insurance, while foreign currency debt provides greater discipline.

Maturity generates an analogous trade-off. Short-term debt limits the government's ability to dilute existing creditors through future borrowing, so lenders are willing to pay more for it. Long-term debt exposes investors to dilution risk because future borrowing can reduce the value of outstanding bonds. At the same time, long-term debt provides insurance because its market value falls when the economy deteriorates and default risk rises. The sovereign therefore owes less in market-value terms in states in which consumption is particularly valuable. Short-term debt provides stronger discipline and greater current borrowing capacity, while long-term debt provides greater insurance against future income risk.

The central contribution of the model comes from combining these two margins. Long-term domestic currency debt combines the insurance properties of both long maturity and domestic denomination. In bad states, inflation can reduce its real repayment value, and higher default risk can lower the market value of its unmatured component. These same characteristics also give it the largest incentive costs: the government has incentives to inflate away both current and future domestic currency obligations, while future borrowing can dilute outstanding long-term creditors' claims. Short-term foreign currency debt lies at the other extreme. It minimizes inflation

and dilution incentives and therefore allows the government to raise revenue on relatively favorable terms, but offers essentially no insurance against bad future states. The other two bonds, short-term domestic currency and long-term foreign currency debt, provide intermediate combinations of insurance and discipline.

These trade-offs generate state-dependent portfolio choice. In periods of severe economic distress, current consumption is particularly valuable and the government places greater weight on raising revenue today. Short-term foreign currency debt becomes attractive because its strong disciplining properties support a relatively high issuance price. Mean reversion also makes its lack of future insurance less costly when current conditions are unusually poor. During economic booms, the government can afford the high incentive costs associated with long-term domestic currency debt and values its insurance against a possible deterioration in future conditions. In the more common intermediate states, neither extreme is attractive enough to offset its corresponding cost, and the government relies more heavily on short-term domestic currency and long-term foreign currency debt.

I quantify the model using Türkiye over 2004–2018, a period of relatively stable growth and inflation. Parameters governing output and the real exchange rate are estimated from Turkish data, while the remaining parameters are chosen to match the debt-to-GDP ratio, the level and volatility of sovereign spreads, and the level and volatility of inflation. The model's implications for currency composition and maturity are left untargeted. It generates a domestic currency share of 81.7 percent, compared with 89.7 percent in the data. More importantly, it reproduces the large maturity difference across currencies. Domestic and foreign currency debt have average maturities of 2.2 and 5.8 years in the model, compared with 3.1 and 8.6 years in the data. Although the model understates both maturity levels, their ratio is close to the data: foreign currency debt is 2.6 times as long as domestic currency debt in the model, compared with 2.8 times in the data.

The quantitative model also reproduces the state dependence of portfolio choice. Short-term foreign currency debt becomes most important when income is low and the domestic currency has depreciated. Long-term domestic currency debt becomes most important when income is high and the domestic currency is strong. Across most of the state space, the two intermediate bonds account for a larger share of issuance. This pattern explains why the unconditional data is characterized by relatively short domestic currency maturity and relatively long foreign currency maturity even though the government occasionally finds the two extreme instruments valuable.

This paper contributes to the literature on sovereign debt management. Following Arellano (2008), a large quantitative literature has studied how debt contracts affect sovereign default incentives. Hatchondo and Martinez (2009) and Chatterjee and Eyigungor (2012) introduce long-term debt, while Arellano and Ramanarayanan (2012), Hatchondo et al. (2016), Bocola and Dovis (2019), and Mihalache (2020) study endogenous maturity choice, generally restricting debt to foreign currency. A parallel literature studies currency denomination, including Aguiar et al. (2013), Ottonello and Perez (2019), Du et al. (2020), Sunder-Plassmann (2020), Lee (2021), Engel and Park (2022), and Maeng (2025). In these environments, maturity or currency is typically varied

while the other dimension is held fixed. The contribution here is to allow the sovereign to choose both dimensions simultaneously and to show that their insurance and incentive properties interact in economically important ways. This joint choice can account for the systematic tendency of emerging-market governments to issue domestic currency debt at shorter maturities than foreign currency debt.

The rest of the paper is organized as follows. First, I present novel empirical evidence on emerging market economies' sovereign debt issuance in Section II. I then present the model and discuss its optimality conditions in Section III and Section IV, respectively. Section V then analyzes the model's quantitative predictions relative to the Turkish economy. Finally, I conclude in Section VI.

II Empirical Analysis

In this section, I analyze data on sovereign bond issuance and spreads. First, I robustly find that sovereign debt denominated in a country's domestic currency tends to be of significantly shorter maturity than debt denominated in a foreign currency. I then find that in times of economic distress, or high spreads, sovereigns tilt their issuance toward shorter maturities and toward more foreign currency debt.

II.A Data Description

I use three datasets in my empirical analysis. The first, which I collect from Bloomberg, covers sovereign bond issuance. It spans 19 emerging market economies from 1990 — when Bloomberg issuance coverage becomes broad across countries — through the end of 2024, an aggregate sample of 24,910 individual bond issuances. For each issued bond, I observe the amount issued, the currency denomination, its date of issuance, and its maturity date. I use issuance data rather than debt stocks because no publicly available stock data is granular enough to decompose debt by currency denomination and maturity simultaneously.[1]

My second and third datasets, from Datastream and Du and Schreger (2016), provide sovereign bond spreads. Specifically, I define the spread as

$$\text{Spread}_{i,n,t} = y^{\text{domestic}}_{i,n,t} - y^{\text{US Treasury}}_{n,t}, \tag{1}$$

where $y^{\text{domestic}}_{i,n,t}$ is the n-year domestic currency bond yield for emerging market economy i at time t, and $y^{\text{US Treasury}}_{n,t}$ is the n-year US Treasury bond yield at time t. Taking the US Treasury yield as the risk-free benchmark, this differential is the premium international investors demand to lend to economy i in its own currency. Higher spreads reflect investors perceiving the sovereign's debt as riskier, whether from default risk, currency risk, or both.

[1] Arslanalp and Tsuda (2012) provide quarterly-frequency data on foreign holdings of sovereign debt but do not provide a breakdown by maturity.

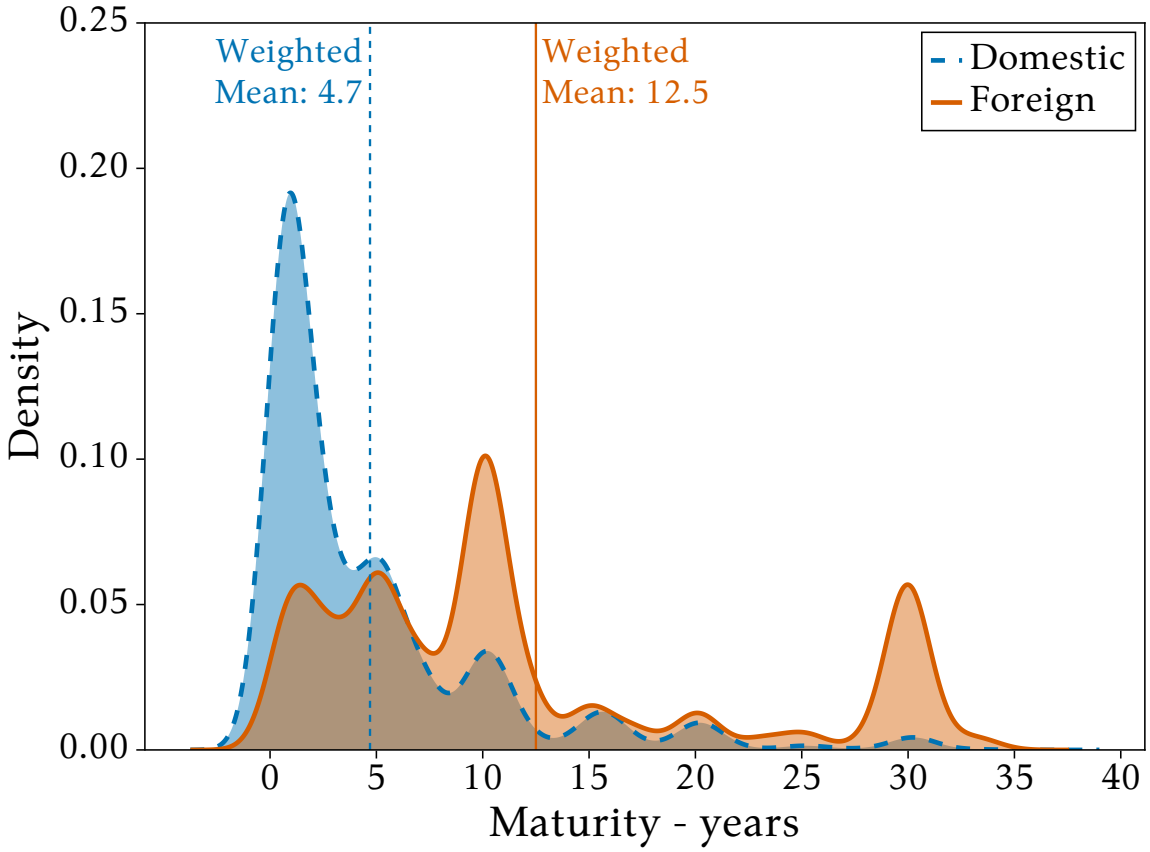


Figure 1: Distribution of Maturity by Currency

Notes: Kernel density estimates of sovereign bond maturity at issuance, by currency denomination, using Bloomberg issuance data for 19 emerging-market sovereigns over 1990–2024. "Domestic" bonds are denominated in the issuer's own currency; "foreign" bonds are denominated in another currency. Densities are weighted by the US-dollar amount issued and estimated with a Gaussian kernel (bandwidth = 1 year); issuances with maturity above 35 years are excluded for visual clarity. Vertical lines denote amount-weighted mean maturities.

I provide summary statistics of each dataset and details on my data cleaning procedures in Appendix A.

II.B Average Maturity by Currency

First, I present the distribution of maturities of all domestic currency and foreign currency denominated bonds in my sample in Figure 1. These density plots show that there is a clear difference in the maturity distributions of domestic and foreign currency debt. Domestic currency debt is heavily concentrated at shorter maturities, with the majority of domestic currency debt having a maturity below 5 years. In contrast, foreign currency debt is concentrated at longer maturities, with the majority of foreign currency debt's maturity being above 10 years.

Next, I divide my data across countries and show the average maturity of sovereign debt

	Weighted		Unweighted	
Country	**Domestic**	**Foreign**	**Domestic**	**Foreign**
Angola	3.4 years	10.6 years	2.7 years	5.1 years
Argentina	6.5 years	13.5 years	3.5 years	8.3 years
Colombia	11.2 years	15.8 years	4.5 years	10.3 years
Egypt	1.6 years	6.6 years	1.5 years	5.6 years
Hungary	4.7 years	10.6 years	2.2 years	8.3 years
Indonesia	10.0 years	14.0 years	4.6 years	13.0 years
Jamaica	8.8 years	6.1 years	6.4 years	7.9 years
Jordan	3.7 years	6.8 years	3.3 years	7.0 years
Mexico	3.8 years	18.4 years	2.9 years	11.7 years
Philippines	5.2 years	15.1 years	2.5 years	11.8 years
Poland	5.6 years	11.6 years	2.3 years	11.5 years
Qatar	4.9 years	15.3 years	5.1 years	13.5 years
Romania	4.8 years	10.2 years	1.6 years	8.6 years
Russia	6.7 years	17.5 years	5.7 years	11.4 years
Tunisia	5.7 years	9.5 years	2.6 years	18.8 years
Türkiye	3.5 years	7.5 years	3.0 years	6.0 years
Ukraine	5.7 years	6.6 years	3.3 years	4.2 years
Uruguay	11.6 years	18.0 years	8.5 years	8.4 years
Venezuela	2.6 years	17.2 years	1.7 years	15.8 years
Average	5.8 years	12.2 years	3.6 years	9.9 years
Median	5.2 years	11.6 years	3.0 years	8.6 years
Standard Deviation	2.8 years	4.3 years	1.8 years	3.8 years

Notes: Average maturity at issuance (in years) of sovereign bonds by currency denomination, using Bloomberg issuance data for 19 emerging-market sovereigns over 1990–2024. "Weighted" columns weight each bond by its amount issued; "unweighted" columns are simple averages across issuances. Country rows average over all bond issuances within that country. The Average, Median, and Standard Deviation rows are computed across the 19 country-level values, weighting each country equally rather than pooling across individual bonds. A paired t-test across the 19 countries rejects equality of the issuance-weighted mean domestic and foreign currency maturities: the average within-country gap (domestic minus foreign) is −6.4 years ($t_{18} = -6.5$, $p < 0.0001$).

Table 1: Average Maturity of External Sovereign Debt by Currency

by currency denomination for each country in my empirical sample in Table 1. The weighted columns show the weighted average maturity of each country's domestic and foreign currency debt, respectively, where each bond is weighted by the amount issued. The unweighted columns show the simple average maturity. Table 1 shows that the maturity gap across currencies holds at the individual country level too, with Jamaica being the only exception.

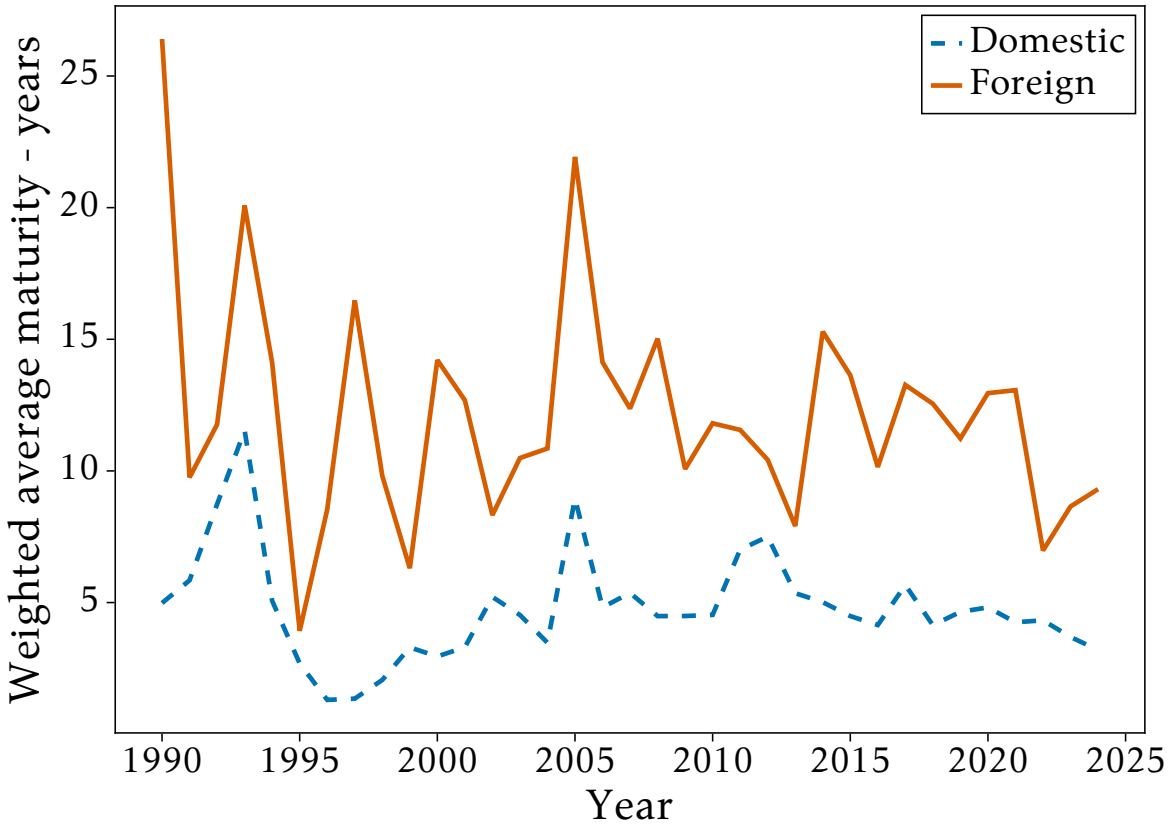


Figure 2: Time Series of Weighted Average Maturity by Currency

Notes: Issuance-amount-weighted average maturity at issuance of sovereign bonds by currency denomination, plotted by year of issuance, using Bloomberg issuance data for 19 emerging-market sovereigns over 1990–2024. For each year, average maturity is computed across all bonds issued that year, pooling across countries, with each bond weighted by its amount issued. Domestic currency bonds are denominated in the issuer's own currency; foreign currency bonds are denominated in another currency.

Lastly, this pattern is also robust across time. I plot the time series of the weighted average maturities of domestic and foreign currency debt in Figure 2. Across every year in my 35-year sample, foreign currency debt has been of longer maturity than domestic currency debt.

II.C Issuance Choices and Spreads

I now turn to the relationship between spreads and the two margins of issuance. Since sovereigns issue debt infrequently relative to the daily frequency of the spread data in Du and Schreger (2016), I aggregate to the quarterly frequency (the frequency of the Datastream data) so that each country-quarter contains enough issuance to define the currency share and average maturity. All subsequent analysis is quarterly unless otherwise specified.

The first column of Table 2 reports the correlation between spreads and the share of debt

Country	Domestic Share	Maturity (All)	Maturity (Domestic)	Maturity (Foreign)
Colombia	-0.090	-0.055	0.016	-0.388
Egypt	-0.386	0.241	0.136	-0.164
Hungary	-0.091	-0.064	-0.085	0.132
Indonesia	-0.135	-0.051	-0.125	-0.117
Jamaica	-0.136	0.047	-0.023	0.065
Mexico	-0.091	-0.124	-0.097	-0.195
Philippines	-0.147	-0.185	-0.179	-0.436
Poland	0.169	-0.350	-0.318	-0.123
Romania	-0.042	-0.323	-0.392	-0.314
Russia	-0.129	-0.274	-0.306	-0.031
Türkiye	-0.224	-0.067	-0.032	-0.277
Uruguay	-0.384	-0.172	-0.055	-0.275

Notes: Each cell reports the within-country Pearson correlation, across country-quarters, between the sovereign's spread and a measure of its debt issuance. "Domestic Share" is the amount-weighted share of a quarter's issuance denominated in the country's own currency. "Maturity (All)," "(Domestic)," and "(Foreign)" are the amount-weighted average maturity (in years) of, respectively, all debt, own-currency debt, and foreign currency debt issued that quarter. Spreads are quarterly averages: for Colombia, Hungary, Indonesia, Mexico, Philippines, Poland, Russia, and Türkiye they are the 3-month local-currency sovereign spread of Du and Schreger (2016); for Egypt, Jamaica, Romania, and Uruguay they are sovereign spreads from Datastream. Correlations use all quarters in which both series are observed; the sample covers the 12 countries with available spread data.

Table 2: Correlation between Spreads and Currency Share and Maturity of Issuance

issued in a country's domestic currency for each quarter across each country. With the exception of Poland, I find that higher spreads are associated with sovereigns tilting their issuance toward more foreign currency debt.

Next, I consider the relationship between spreads and the maturity of debt issued. The next three columns of Table 2 show the correlation between spreads and the amount-weighted average maturity of all debt, domestic currency debt, and foreign currency debt, respectively. I find that as spreads increase, sovereigns shorten the maturity of their debt issuance. While this holds for domestic currency debt across most countries, it is stronger for foreign currency debt.

Next, I report the results of several univariate regressions of various measures of the sovereign's portfolio choice on spreads with country and year fixed effects. Specifically, Column (1) of Table 3 estimates, on the country-quarter panel,

$$\text{Domestic Share}_{c,q} = \beta\,\text{Spread}_{c,q} + \alpha_c + \delta_{y(q)} + \varepsilon_{c,q},$$

where $\text{Domestic Share}_{c,q}$ is the amount-weighted share of debt issued by country c in quarter q that is denominated in its own currency, $\text{Spread}_{c,q}$ is the quarterly-average spread, $y(q)$ is the year

	(1)	(2)	(3)	(4)
	Domestic Share	Maturity (Years)		
		All	Domestic	Foreign
Spread	−0.003***	−0.054**	−0.050**	−0.097*
	(0.001)	(0.022)	(0.023)	(0.053)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,077	1,077	1,016	560
R^2	0.259	0.271	0.248	0.245

Notes: Each column reports an OLS regression on the country-quarter panel of a measure of the sovereign's debt issuance on the quarterly-average spread. The dependent variable in Column (1) is the amount-weighted share of a quarter's issuance denominated in the country's own currency ("Domestic Share"). In Columns (2)–(4) it is the amount-weighted average maturity (in years) of all debt, own-currency debt only, and foreign currency debt only, respectively. All regressions include country and year fixed effects. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 3: Association between Spreads and Portfolio Composition

containing quarter q, and α_c, δ_y are country and year fixed effects. Columns (2)–(4) estimate, on the same panel,

$$\text{Maturity}^k_{c,q} = \beta^k \, \text{Spread}_{c,q} + \alpha^k_c + \delta^k_{y(q)} + \varepsilon^k_{c,q}, \qquad k \in \{\text{All, Domestic, Foreign}\},$$

where $\text{Maturity}^k_{c,q}$ is the amount-weighted average maturity of debt of type k issued by country c in quarter q.

Column (1) yields an estimate of −0.003, which is statistically significant at the 1% level and indicates that a one-percentage-point increase in the spread is associated with a 0.34 percentage-point decline in the share of a quarter's issuance denominated in the sovereign's own currency (relative to a sample mean of 81%), controlling for country and year fixed effects. Columns (2)–(4) show that a one-percentage-point increase in spreads is associated with the maturity of all debt issued decreasing by 0.054 years (or approximately 3 weeks). If only debt denominated in foreign currency is considered, then the associated drop increases to approximately 5 weeks.

Lastly, I report the results of a logit regression of an indicator that equals one if a bond is denominated in the issuing country's own currency and its maturity is strictly above the median maturity of all debt (both currencies) issued by that country over the sample, on one-day-lagged spread. Since this regression is estimated at the bond-issuance level, I include only the eight countries in the Du and Schreger (2016) dataset since the Datastream data is only available at the quarterly frequency. I use the lagged spread instead of the contemporaneous spread in order to ameliorate simultaneity bias. Furthermore, I use domestic currency long-term debt instead of foreign currency short-term debt for two reasons. First, Du and Schreger (2016)'s spread data is

	(1)
	$\mathbb{1}\{$Domestic Currency, Long Maturity$\}$
	(Logit)
Spread	-0.025^{***}
	(0.008)
Country FE	Yes
Year FE	Yes
Observations	7,982
R^2	–

Notes: The table reports a logit regression, estimated at the bond-issuance level, of an indicator that equals one if a bond is denominated in the issuing country's own currency and its maturity is strictly above the median maturity of all debt (both currencies) issued by that country over the sample, on the spread lagged by one day. The spread is the 3-month local-currency sovereign spread of Du and Schreger (2016), measured on the issuance date (or the nearest prior available date). The regression includes country and year fixed effects and covers the eight countries in the Du-Schreger data: Colombia, Hungary, Indonesia, Mexico, Philippines, Poland, Russia, and Türkiye. Standard errors in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 4: Association between Spreads and Long-Maturity Domestic Currency Issuance

on domestic currency debt. Second, foreign currency debt is issued significantly less frequently than domestic currency debt.

The results are reported in Table 4. The coefficient on spreads is negative and statistically significant at the 1% level, indicating that higher spreads are associated with a lower probability of issuing long-maturity domestic currency debt.

II.D Summary

In summary, I use microdata on bonds issued by emerging market economies and on spreads to document the following three stylized facts. First, sovereign debt denominated in a country's domestic currency tends to be of much shorter maturity than its debt denominated in a foreign currency. Second, in times of economic distress, sovereigns are more likely to issue debt denominated in a foreign currency. Lastly, sovereigns also lower the maturity of all their issued debt during times of crisis, with this effect being more pronounced among foreign currency debt.

The first stylized fact is new to the literature and has two important implications. First, its robustness across both time and countries suggests that when deciding what types of debt to issue, emerging market economies do consider both portfolio margins jointly rather than in isolation. Second, short-term foreign currency bonds and long-term domestic currency bonds must have some economic costs that make them unattractive to issue during normal times.

My second and third stylized facts extend the existing analysis in the sovereign debt literature. Ottonello and Perez (2019) show that the domestic currency share of debt is procyclical with

respect to the business cycle, while Maeng (2025) finds that the stock of domestic currency debt is negatively related to a sovereign's borrowing costs. Arellano and Ramanarayanan (2012) and Broner et al. (2013) document that higher spreads are associated with shorter maturities of debt issued, but they only consider debt denominated in foreign currency.

The fact that sovereigns tilt their issuance portfolio toward more short-term, foreign currency debt during periods of high risk suggests that short-term, foreign currency debt has benefits that render issuing it optimal during times of economic distress, while, conversely, long-term, domestic currency debt is only attractive during economic booms.

III Model

In this section, I outline an infinite-horizon, discrete-time, dynamic model of a small open economy borrowing from foreign lenders. There is a benevolent government that aims to maximize the welfare of a representative risk-averse and hand-to-mouth household by choosing the currency and maturity composition of its debt and inflation. The government can only borrow from risk-neutral and perfectly competitive international investors. Crucially, the government cannot commit to what it will do in the future regarding (i) whether it will repay its debts, (ii) how it will borrow, and (iii) how it will conduct monetary policy.

III.A Environment

III.A.1 Households

There is a representative risk-averse household whose preferences are given by:

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\left[u(C_t)-l(\Pi_t-\Pi^*)\right]\right], \tag{2}$$

where $\beta\in(0,1)$ is the household's discount factor, C_t is consumption, Π_t is inflation, and Π^* is the target inflation rate. The household receives utility from consumption as given by the strictly increasing and concave utility function $u:\mathbb{R}_+\to\mathbb{R}$, and disutility from inflation that deviates from the target inflation rate as given by the disutility function $l(x)=\psi\,\ell(x)$, where $\psi>0$ governs the household's sensitivity to inflation deviations from target and $\ell:\mathbb{R}\to\mathbb{R}_+$ is convex, strictly increasing on $\mathbb{R}_+$, and satisfies $\ell(0)=0$.[2]

The economy receives a stochastic endowment of the domestic consumption good, $\mathcal{Y}_t$, which differs from the international consumption good that the foreign lenders consume. I define $\mathcal{E}_t$ as the number of real domestic consumption goods that can be purchased with one unit of the

[2] In principle, $\Pi_t-\Pi^*$ can be negative, in which case ℓ need not be increasing. In equilibrium, however, the government never chooses inflation below target, since inflation only benefits it by eroding the real value of its domestic currency debt (see (19)). Only the increasing branch of ℓ on $\mathbb{R}_+$ is therefore relevant.

international consumption good in period t.[3] Lastly, I assume that the household is hand-to-mouth: it consumes the lump-sum transfer T_t that it receives from the government in each period, so that $C_t = T_t$ whenever the government repays.

III.A.2 Government

The government is benevolent and aims to maximize the representative household's utility. Every period, the government observes the realized value of its endowment and real exchange rate, denoted by $\mathbb{X}_t = (\mathcal{E}_t, \mathcal{Y}_t)$. The government then decides whether to default on all its debt obligations that are due in period t or repay them.

If the government decides to default, then all its debt obligations are completely erased. However, following Arellano (2008), there are two costs to defaulting. First, the government will be excluded from international financial markets, and therefore unable to borrow from international investors, for a stochastic number of periods; in each period of exclusion, it regains access with probability θ, independently over time. Second, there are output costs to defaulting, which are given by $h(\mathcal{Y}_t)$. The household's consumption under default is then given by:

$$C_t = h(\mathcal{Y}_t). \tag{3}$$

If the government chooses to repay its period t debt obligations, it can finance repayment and transfers to the household, T_t, with a combination of (i) issuing new debt and (ii) inflation. If the government owes any debt denominated in its domestic currency, then it can use inflation to dilute the real cost of repaying its domestic currency debt, albeit at the cost of increasing disutility to the household.

In addition, the government can issue four types of bonds: (i) short-term, domestic currency bonds (b^{SD}_{t+1}), (ii) short-term, foreign currency bonds (b^{SF}_{t+1}), (iii) long-term, domestic currency bonds (b^{LD}_{t+1}), and (iv) long-term, foreign currency bonds (b^{LF}_{t+1}) at prices q^{SD}, q^{SF}, q^{LD}, and q^{LF}, respectively. These prices are all endogenous to the government's present borrowing choices and current state of the economy, since they all jointly determine the government's incentives in future periods and thus the risks that lenders face.

Issuing the short-term, foreign currency bond is a contract in which, conditional on not defaulting, the government must pay b^{SF}_{t+1} units of the foreign good in period $t+1$ in exchange for receiving $q^{SF} b^{SF}_{t+1}$ units of the foreign good in period t, which is worth $\mathcal{E}_t q^{SF} b^{SF}_{t+1}$ units of the domestic good. I model long-term debt in the same manner as Hatchondo and Martinez (2009), where for each unit of long-term foreign currency debt b^{LF}_{t+1}, a fraction $\delta \in (0,1]$ matures in each period. Hence, the law of motion for the long-term, foreign currency bond is given by:

$$i^{LF}_t = b^{LF}_{t+1} - (1-\delta) b^{LF}_t, \tag{4}$$

where i^{LF}_t is the amount of long-term, foreign currency debt that is issued in period t and $(1-\delta)b^{LF}_t$ denotes the amount of unmatured, long-term foreign currency debt.

[3] $\mathcal{E}_t$ increasing corresponds to the domestic currency depreciating.

Short-term and long-term domestic currency bonds are modeled in the same manner as their foreign currency counterparts, except that the units of repayment are denominated in the domestic good instead of the foreign good.

Altogether, the government's period t resource constraint (in the domestic consumption good) is given by

$$T_t + \mathcal{E}_t\left[b_t^{SF} + b_t^{LF}\right] + \frac{b_t^{SD} + b_t^{LD}}{\Pi_t} = q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SD} + q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)\left[b_{t+1}^{LD} - (1-\delta)\frac{b_t^{LD}}{\Pi_t}\right] \tag{5}$$
$$+\mathcal{E}_t\left[q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SF} + q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t)\left[b_{t+1}^{LF} - (1-\delta) b_t^{LF}\right]\right] + \mathcal{Y}_t,$$

where the left-hand side of the first line represents the government's expenditures (transfers and repaying its debts), the right-hand side of the first line represents the government's revenues from its domestic currency bonds, and the second line represents the government's revenues from its foreign currency bonds and its endowment.

For notational convenience, let $\mathbb{B}_t = (b_t^{SD}, b_t^{SF}, b_t^{LD}, b_t^{LF})$ denote the vector of period t debt obligations for each type of bond.

III.A.3 Foreign Lenders

The government's debt can only be purchased by foreign lenders, who I assume are risk-neutral and perfectly competitive, resulting in them breaking even in expectation. These investors can access a riskless security that yields a constant gross interest rate of R^*.

In order to induce an investor to purchase the government's debt, the investor must be compensated for the risks that holding each bond exposes them to. The short-term foreign currency bond only needs to compensate investors for default risk, since its real value is exogenous to the government's borrowing and monetary policy choices. In contrast, the government can manipulate the real value of its domestic currency debt via inflation, thereby exposing investors to inflationary risk if they lend to the government in its domestic currency.

Maturity introduces a further source of risk. Since long-term bonds are repaid to investors over multiple periods, holding them exposes investors to an additional source of risk relative to short-term bonds: dilution risk, as the government debt literature calls it. Due to the government's inability to commit to future borrowing (and monetary policy), investors are concerned that the government may issue new debt in the future, which would dilute the real value of their unmatured debt holdings.

Altogether, risks accumulate along both margins: the short-term foreign currency bond carries only default risk, while the long-term domestic currency bond carries default, dilution, and inflation risk.

III.B Recursive Problem

As is standard in the literature, I focus on Markov Perfect equilibria and I represent the government's infinite-horizon decision problem as a recursive dynamic programming problem. Each

period, the government observes its aggregate state, which consists of its current debt obligations $\mathbb{B}_t$ and the exogenous state variables $\mathbb{X}_t$. The government then chooses whether to repay or default on its debt obligations. Thus, the government's beginning-of-period value function is given by:

$$V(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{D,R\}} \Big\{ V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t) \Big\}. \tag{6}$$

If the government repays $\mathbb{B}_t$, it chooses the currency and maturity composition of its new borrowing, $\mathbb{B}_{t+1}$, and the level of inflation. The value of repaying is given by:

$$V^R(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{\mathbb{B}_{t+1}, \Pi_t\}} \Big\{ u(C_t) - l(\Pi_t - \Pi^*) + \beta \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[V(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \right] \Big\} \tag{7}$$

subject to the resource constraint (5) and the bond pricing functions.

If the government defaults, then all its debt obligations are permanently erased. In periods of default, the government has no domestic currency debt obligations, and thus has no incentive to deviate from its inflation target rate, Π^*. However, the government will suffer output costs and be in financial autarky for a stochastic number of periods. Together, the value of defaulting is given by:

$$V^D(\mathbb{X}_t) = u(h(\mathcal{Y}_t)) + \beta \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \Big[\theta V(\mathbf{0}, \mathbb{X}_{t+1}) + (1-\theta) V^D(\mathbb{X}_{t+1}) \Big]. \tag{8}$$

Let the government's policy function for repaying be denoted by the indicator function:

$$r(\mathbb{B}_t, \mathbb{X}_t) = \mathbb{1} \Big\{ V^R(\mathbb{B}_t, \mathbb{X}_t) \geq V^D(\mathbb{X}_t) \Big\}. \tag{9}$$

Furthermore, let variables with a hat superscript denote their policy function counterparts. Together, the foreign investors' pricing functions for each type of bond are given by:

$$q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \right], \tag{10}$$

$$q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \Big[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \Big[1 + (1-\delta) q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1}) \Big] \Big], \tag{11}$$

$$q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\hat{\Pi}_{t+1}} \right], \tag{12}$$

$$q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*} \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\hat{\Pi}_{t+1}} \times \Big[1 + (1-\delta) q^{LD}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1}) \Big] \right]. \tag{13}$$

For lenders, all bonds are subject to default risk as given by the government's default policy function $r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1})$. To understand the domestic currency bond pricing functions, note that the short-term, domestic currency bond pays $\frac{1}{\Pi_{t+1}}$ units of the domestic good in period $t+1$, which is worth $\frac{1}{\mathcal{E}_{t+1}\Pi_{t+1}}$ units of the foreign good. Lastly, the aforementioned dilution risk present in all long-term debt is captured by the $q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})$ and $q^{LD}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})$ terms in the long-term bond pricing functions, where $\mathbb{B}_{t+2}$ is given by the government's policy functions for portfolio choice in equilibrium.

Definition 1 (Markov Perfect Equilibrium). *A Markov Perfect recursive equilibrium for this economy consists of a set of:*

(i) value functions $\left\{V(\mathbb{B}_t, \mathbb{X}_t), V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t)\right\}$,

(ii) pricing functions $\left\{q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)\right\}$, *and*

(iii) policy functions $\left\{\hat{b}^{LD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{LF}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SF}(\mathbb{B}_t, \mathbb{X}_t), \hat{\Pi}_t(\mathbb{B}_t, \mathbb{X}_t)\right\}$

such that the following conditions hold:

1. *Taking the bond price functions as given, the government's policy functions and value functions solve the government's dynamic programming problem, and*
2. *The bond price functions* $q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)$ *satisfy* (10), (11), (12), *and* (13), *respectively.*

IV Optimal Portfolio

In this section, I discuss the first-order necessary conditions for each of the four bonds that the government may issue.

This section is organized as follows. First, I present the intertemporal optimality condition of each bond corresponding to the full problem in Section IV.A. Then, in order to show the trade-offs between each bond more transparently, I use the short-term foreign currency bond as the baseline and I compare it against each of the other three bonds in turn. I begin by comparing it against the short-term domestic currency bond in Section IV.B, where the only margin of adjustment is currency denomination. Next, I compare it against the long-term foreign currency bond in Section IV.C, where the only margin of adjustment is maturity. Finally, I shift both margins simultaneously by comparing it against the long-term domestic currency bond in Section IV.D.

IV.A Full Problem

Let $R(\mathbb{B}_t)$ denote the set of $\{\mathcal{E}_t, \mathcal{Y}_t\}$ realizations for which the government chooses to repay its debts for a given vector of obligations $\mathbb{B}_t$:

$$R(\mathbb{B}_t) = \left\{\mathcal{E}_t, \mathcal{Y}_t : V^R(\mathbb{B}_t, \mathbb{X}_t) \geq V^D(\mathbb{X}_t)\right\}. \tag{14}$$

Assuming repayment, the first-order necessary conditions for the government's optimal portfolio choice are given by the following four generalized consumption-Euler equations, which correspond to the first-order conditions of (7) with respect to each bond's issuance decision.

$$u_C(C_t)\left[q^{SF} + \frac{\partial q^{SF}}{\partial b^{SF}_{t+1}} b^{SF}_{t+1} + \frac{\partial q^{LF}}{\partial b^{SF}_{t+1}}\left[b^{LF}_{t+1} - (1-\delta)b^{LF}_t\right] + \frac{\partial q^{SD}}{\partial b^{SF}_{t+1}} b^{SD}_{t+1} + \frac{\partial q^{LD}}{\partial b^{SF}_{t+1}}\left[b^{LD}_{t+1} - (1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \tag{15}$$
$$= \beta \int_{R_{t+1}} u_C(C_{t+1})\mathcal{E}_{t+1} f(\mathbb{X}_t, \mathbb{X}_{t+1}) d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{SD}+\frac{\partial q^{SF}}{\partial b^{SD}_{t+1}}b^{SF}_{t+1}+\frac{\partial q^{LF}}{\partial b^{SD}_{t+1}}\left[b^{LF}_{t+1}-(1-\delta)b^{LF}_t\right]+\frac{\partial q^{SD}}{\partial b^{SD}_{t+1}}b^{SD}_{t+1}+\frac{\partial q^{LD}}{\partial b^{SD}_{t+1}}\left[b^{LD}_{t+1}-(1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \quad (16)$$
$$=\beta\int_{R_{t+1}}u_C(C_{t+1})\frac{1}{\Pi_{t+1}}f(\mathbb{X}_t,\mathbb{X}_{t+1})d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{LF}+\frac{\partial q^{SF}}{\partial b^{LF}_{t+1}}b^{SF}_{t+1}+\frac{\partial q^{LF}}{\partial b^{LF}_{t+1}}\left[b^{LF}_{t+1}-(1-\delta)b^{LF}_t\right]+\frac{\partial q^{SD}}{\partial b^{LF}_{t+1}}b^{SD}_{t+1}+\frac{\partial q^{LD}}{\partial b^{LF}_{t+1}}\left[b^{LD}_{t+1}-(1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \quad (17)$$
$$=\beta\int_{R_{t+1}}u_C(C_{t+1})\left[1+(1-\delta)q^{LF}_{t+1}\right]\mathcal{E}_{t+1}f(\mathbb{X}_t,\mathbb{X}_{t+1})d\mathbb{X}_{t+1}$$

$$u_C(C_t)\left[q^{LD}+\frac{\partial q^{SF}}{\partial b^{LD}_{t+1}}b^{SF}_{t+1}+\frac{\partial q^{LF}}{\partial b^{LD}_{t+1}}\left[b^{LF}_{t+1}-(1-\delta)b^{LF}_t\right]+\frac{\partial q^{SD}}{\partial b^{LD}_{t+1}}b^{SD}_{t+1}+\frac{\partial q^{LD}}{\partial b^{LD}_{t+1}}\left[b^{LD}_{t+1}-(1-\delta)\frac{b^{LD}_t}{\Pi_t}\right]\right] \quad (18)$$
$$=\beta\int_{R_{t+1}}u_C(C_{t+1})\times\frac{1}{\Pi_{t+1}}\times\left[1+(1-\delta)q^{LD}_{t+1}\right]f(\mathbb{X}_t,\mathbb{X}_{t+1})d\mathbb{X}_{t+1}$$

The left-hand side of each equation corresponds to the marginal benefit of issuing an additional unit of that bond, while the right-hand side corresponds to the marginal cost of issuing an additional unit of that bond. The marginal benefit is given by the marginal utility of consumption multiplied by the marginal revenue from issuing an additional unit of that bond, which is captured by the terms in the square brackets. This marginal revenue includes the bond's own price, as well as the effect of additional issuance of that bond on its own price and on the prices of all other bonds. Since the incentive to default and inflate next period increases with all debt issuance, all bond prices decrease with respect to issuance. The marginal cost, in turn, is given by the discounted expected marginal utility of consumption in the next period, adjusted for inflation and any future bond prices.

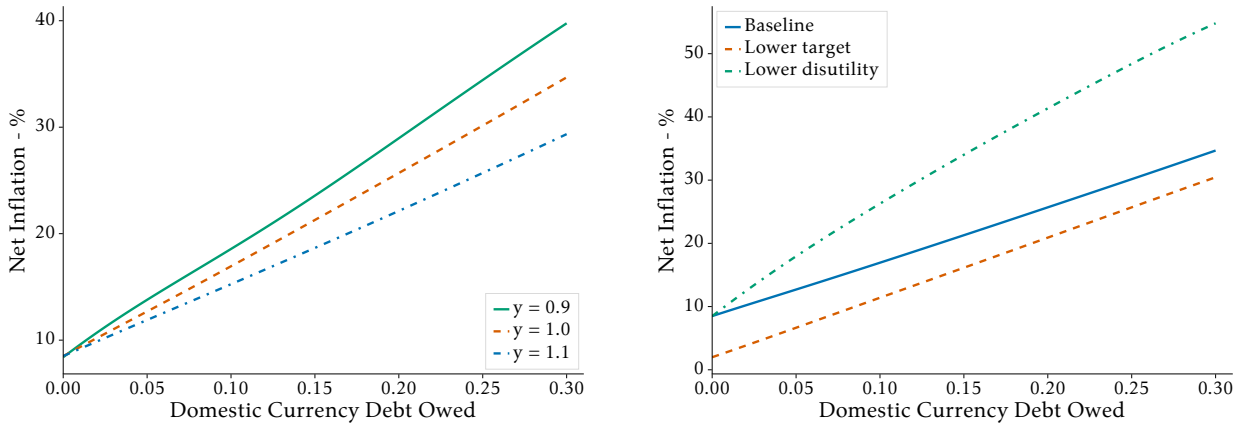
IV.B Currency Choice

In this subsection, I highlight the economic trade-offs between domestic and foreign currency debt, where both bonds are short-term. I first show that issuing foreign currency debt has disciplining benefits for the government, and then present domestic currency debt's hedging benefits.

In a simplified model where all debt is short-term, the government's intratemporal optimality condition for inflation is:

$$u_C(C_t)\left[\frac{1}{\Pi^2_t}\right]\left[b^{SD}_t\right]=l_\Pi(\Pi_t-\Pi^*). \quad (19)$$

As long as the government owes any amount of domestic currency debt, strictly positive net inflation will dilute the real cost of repaying that debt, thereby relaxing the government's resource constraint and allowing higher transfers to the household for consumption. The government's



(a) Inflation Policy Function

(b) Inflation Policy Function Comparative Statics

Figure 3: Policy Function for Inflation

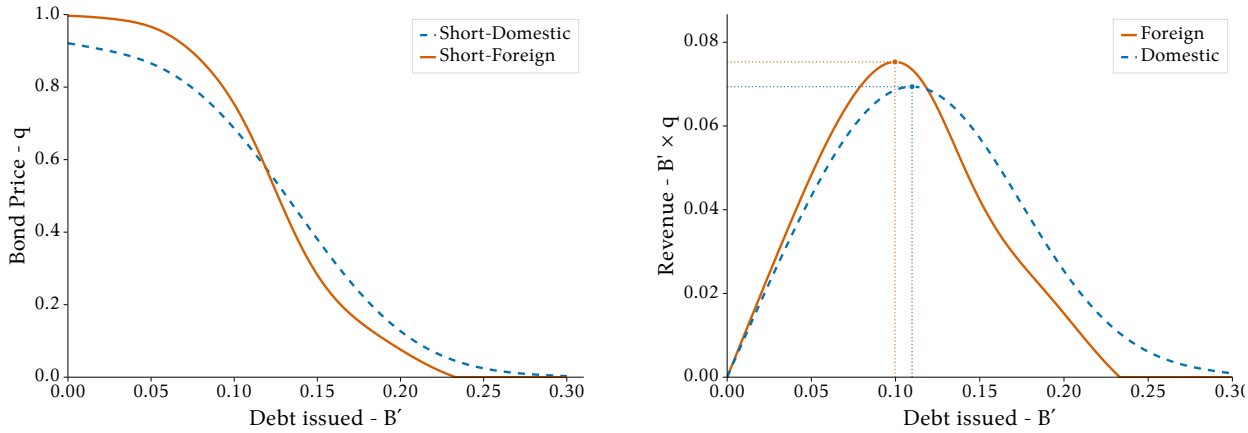
Notes: Both panels plot the equilibrium net inflation policy function (in percent) as a function of domestic currency debt owed, holding foreign currency debt at zero. Panel (a) fixes $\Pi^* = 1.083$ and $\psi = 1.1$ and varies the endowment realization $\mathcal{Y} \in \{0.9, 1.0, 1.1\}$. Panel (b) fixes the endowment at $\mathcal{Y} = 1.0$ and varies the target inflation rate and inflation disutility parameter one at a time relative to the baseline: "Lower target" sets $\Pi^* = 1.02$; "Lower disutility" sets $\psi = 0.4$.

choice of inflation equates the marginal utility gain of additional consumption (the left-hand side of (19)) with the marginal disutility from deviating from its inflation target rate Π^* (the right-hand side of (19)).

Figure 3a shows the government's equilibrium policy function for inflation across three different income levels while varying the amount of domestic currency debt owed on the x-axis. The more domestic currency debt owed, the more the government is tempted to choose a level of inflation that is higher than its target inflation rate. Lenders anticipate this behavior and must be compensated for inflationary risk next period if they are to hold domestic currency debt instead of foreign currency debt.

In order to understand the economic role of the government's target inflation rate, Π^*, and disutility parameter, ψ, I plot the different inflation policy functions across different parameter values (holding the state constant) in Figure 3b. When the government's target inflation is lower, the policy function shifts downward, since the government starts from a lower level of inflation at any amount of debt owed. The shift is not parallel, however: the gap between the two policy functions narrows as domestic currency debt owed increases. With a higher target, inflation is higher at every level of debt, so the marginal utility gain from additional inflation, which falls with the level of inflation through the $1/\Pi_t^2$ term in (19), rises more slowly with domestic currency debt owed, resulting in a flatter policy function.

Next, if the government's disutility parameter is lower, optimal inflation at $b_t^{SD} = 0$ is unchanged since there is no consumption motive to deviate from the target. However, as the amount of domestic currency debt owed increases, the marginal disutility from inflation increases more



(a) Baseline Pricing Functions: Currency

(b) Baseline Revenue Curves: Currency

Figure 4: Incentive Costs of Domestic Currency Debt

Notes: Panel (a) plots the equilibrium price schedule for one-period bonds as a function of debt issued next period (B'), holding the other currency's debt at zero and fixing the exogenous state at $\mathcal{Y} = 1.1$, $\mathcal{E} = 1$. Panel (b) plots implied revenue, $B' \times q(B')$; dotted lines and markers indicate the revenue-maximizing debt level for each currency. "Short-Domestic"/"Short-Foreign" denote one-period bonds denominated in domestic and foreign currency, respectively. Target inflation is set to $\Pi^* = 1.083$ (8.3% net) and the inflation disutility parameter to $\psi = 1.1$.

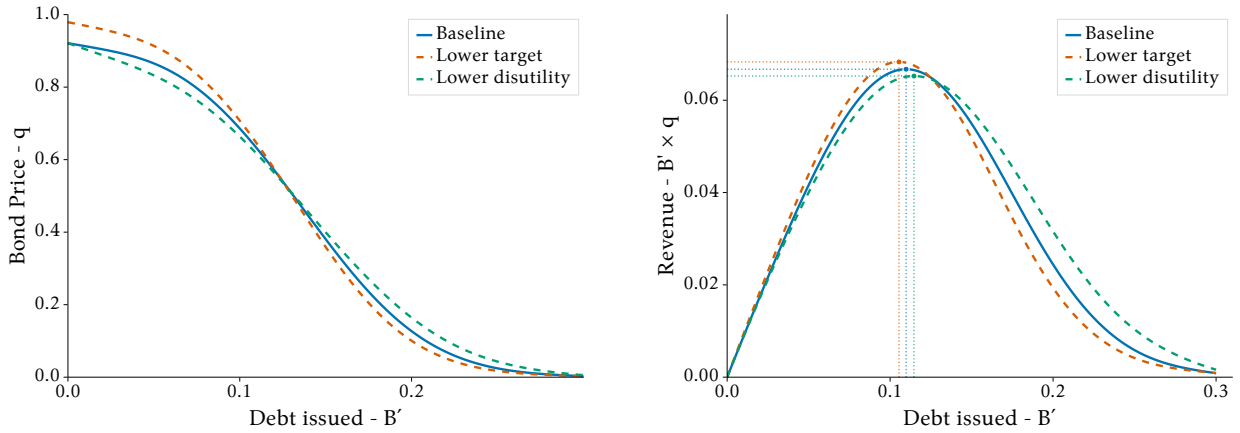
slowly, resulting in a steeper policy function.

The effects of these forces can be seen in Figure 4a, which shows the pricing functions for both domestic and foreign currency bonds as a function of the respective amount of debt issued while holding the other bond issuance constant at zero. There are two noteworthy aspects. First, the initial wedge between the two pricing functions reflects the government's target inflation rate, which acts as a haircut that investors incur with certainty, irrespective of what state the government is in next period.

Second, the foreign currency bond pricing function is steeper than that of the domestic currency bond. As reflected in the negative slope of both pricing functions, the government's incentive to default rises with the amount of debt issued, irrespective of currency denomination. However, default risk rises more quickly with respect to foreign currency than domestic currency debt. As long as the government targets a strictly positive net inflation rate, then it can sustain a higher amount of domestic currency debt before it would find it optimal to default in equilibrium.

While issuing more domestic currency debt increases the government's incentive to default and engage in costly inflation, these incentive costs grow more slowly with domestic currency debt issuance than with foreign currency debt issuance. As a result, the initial premium the government pays from the target inflation rate is overtaken by default risk growing faster with foreign currency debt issuance.

The relative disciplining benefits of foreign currency debt over domestic currency debt can be seen in the revenue curves in Figure 4b. I vary the amount of domestic and foreign currency debt



(a) Pricing Functions: Currency Choice

(b) Revenue Curves: Currency Choice

Figure 5: Comparative Statics: Currency Choice

Notes: Both panels plot the domestic currency bond only, holding foreign currency debt at zero and fixing the exogenous state at $\mathcal{Y} = 1.1$, $\mathcal{E} = 1$, as in Figure 4. Panel (a) plots the equilibrium price schedule $q(B')$; panel (b) plots implied revenue, $B' \times q(B')$, with dotted lines and markers indicating the revenue-maximizing debt level for each parameterization. "Baseline" sets target inflation $\Pi^* = 1.083$ and the inflation disutility parameter $\psi = 1.1$. "Lower target" lowers target inflation to $\Pi^* = 1.02$. "Lower disutility" lowers the disutility parameter to $\psi = 0.4$.

issued along the x-axis (while holding the other bond constant at zero), and plot the corresponding revenue generated from issuing that debt along the y-axis. For both bonds, revenue initially increases with debt issuance. However, due to the negative slope of the pricing function, the revenue eventually peaks and then decreases with more debt issuance since the negative price effects dominate. Not only can the government raise more revenue from issuing foreign currency debt than domestic currency debt, but it is also able to accomplish that higher level of revenue with a lower amount of debt issued and thus owed next period.

Note that the revenue functions and pricing functions simply reflect the government's choice set for a given state of the world. In other words, they reflect what options the government can choose among, but which portfolio it ultimately chooses from that menu in equilibrium is further determined by the government's debt burden.

Figure 5a shows the equilibrium domestic currency pricing function for different values of the government's target inflation rate and inflation disutility parameter. When the target rate is lower, the initial wedge shrinks, while a lower inflation disutility parameter leaves the initial price unchanged but renders the pricing function initially steeper, i.e., its negative slope is larger in magnitude. This is because the less disutility the government incurs from inflation, the more it is tempted to inflate away its domestic currency debt, so the incentive costs of that debt grow faster with issuance.

Lastly, the effects of the government's target inflation rate and inflation disutility parameter on revenue generation are shown in Figure 5b. When the target inflation rate is lower, the government needs to issue less domestic currency debt to attain its maximum, while lowering the inflation

disutility parameter increases the required amount of debt issuance. Meanwhile, how varying the two parameters affects the maximum revenue further depends on the government's value function for defaulting, and especially the output costs from defaulting.

In this example, the government's output costs from defaulting are relatively low, so default risk increases faster with debt issued, and the larger initial wedge from the higher (baseline) target inflation rate dominates the flatter pricing function it generates; as a result, lowering the target raises maximized revenue. However, if the government's output costs from defaulting were higher, the default risk would grow more slowly, and a lower target inflation rate could result in a lower maximized revenue. Lowering the disutility parameter, which steepens the pricing function at the same initial price, lowers maximized revenue.

On the other hand, domestic currency debt provides the government with hedging benefits. This can be seen from the right-hand side of (15) and (16), which are the generalized consumption-Euler equations for the foreign and domestic currency bonds, respectively. A simple covariance decomposition of the right-hand side of (15) shows that the marginal cost of repaying one unit of the foreign currency bond next period depends on the covariance between next period's real exchange rate and the household's marginal utility of consumption. For most emerging market economies, the real exchange rate is negatively correlated with real income. This means that periods of low endowment realizations, when marginal utility tends to be high, are associated with the government's domestic currency depreciating relative to the foreign currency. Thus, the real cost of repaying any foreign currency bond next period is highest precisely when consumption is most valuable.

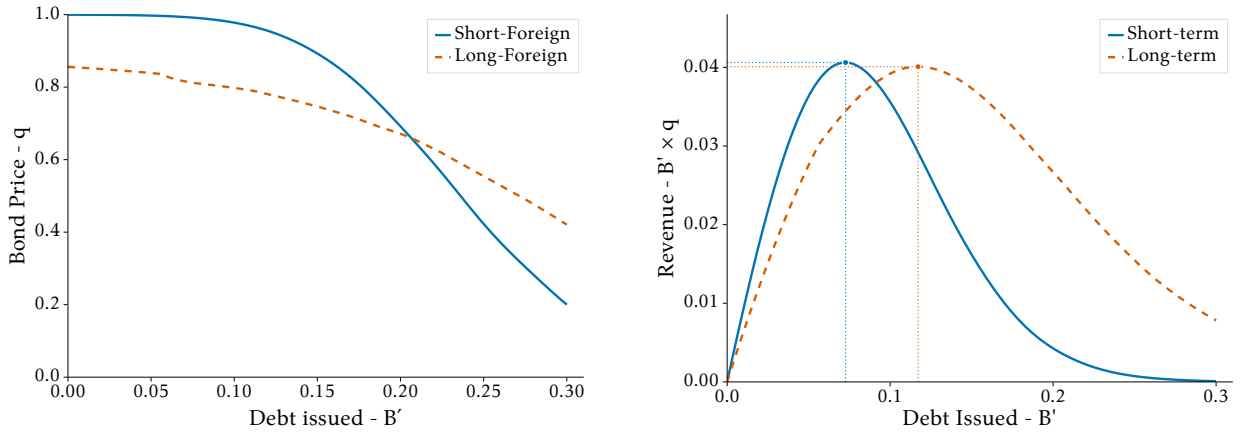
Meanwhile, the real cost of servicing domestic currency debt next period depends on the inverse of the government's choice of next period's inflation rate. Note how the government's policy function for inflation in Figure 3a is downward sloping with respect to income for the same level of domestic currency indebtedness. This means that the government would choose countercyclical inflation; in periods of low income when marginal utility of consumption is high, the government has the flexibility to choose a higher level of inflation, which reduces the real cost of repaying the debt.

IV.C Maturity Choice

I now highlight the disciplining-hedging trade-off between short-term and long-term debt. Since the margin of adjustment is now maturity, the analysis in this subsection assumes that all debt is denominated in foreign currency, and that the real exchange rate is constant and at unity.[4]

Long-term debt's incentive cost relative to short-term debt is illustrated by the additional $(1-\delta)q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})$ term on the right-hand side of (17) relative to (15), which reflects the market

[4]Where possible, the parameter values for this exercise mirror those used in Arellano and Ramanarayanan (2012). The only exception is that I use the output cost function from Chatterjee and Eyigungor (2012), which I outline in Section V.A. I set $\lambda_0 = -0.51$ and $\lambda_1 = 0.53$, which approximately resemble the threshold-cost specification in Arellano and Ramanarayanan (2012).



(a) Pricing Functions: Maturity Choice

(b) Revenue Curves: Maturity Choice

Figure 6: Incentive Costs of Long-Term Debt

Notes: Panel (a) plots the equilibrium price schedule as a function of debt issued next period (B'), holding the other maturity's debt at zero and fixing the exogenous state at $\mathcal{Y} = 1.1$, $\mathcal{E} = 1$. Panel (b) plots implied revenue, $B' \times q(B')$; dotted lines and markers indicate the revenue-maximizing debt level for each maturity. "Short-Foreign"/"Long-Foreign" denote one-period and geometric-decay perpetuity ($\delta = 0.2$) bonds, respectively, both denominated in foreign currency with the real exchange rate fixed at unity.

value of next period's unmatured, long-term debt. This term arises because the government lacks commitment with regard to its future borrowing choices. In period t, the government may choose a b^{LF}_{t+1} such that default risk in period $t+1$ is low, but it cannot commit to not choosing a b^{LF}_{t+2} that increases default risk in period $t+2$, thereby lowering the market value of the unmatured long-term debt. This is in contrast to short-term debt, which matures in one period and does not have any future market value. This risk, which the maturity choice literature often describes as dilution risk, increases with the maturity of the long-term bond: the longer the maturity, the more its holders are exposed to it.

Forward-looking investors anticipate this dilutionary incentive and factor it into their pricing, resulting in the short-term and long-term price schedules as shown in Figure 6a. Both price schedules are evaluated at the same exogenous state, $\mathcal{Y} = 1.1$, $\mathcal{E} = 1$, and with the other maturity's debt issuance held at zero. In such a state, next period's default risk is low, which is why the short-term bond's price immediately begins at the risk-free price. However, due to the aforementioned dilutionary incentive, the long-term bond's price is lower than the short-term bond's price at low levels of debt issuance.

These price schedules translate to the two revenue curves in Figure 6b, which show the revenue curves for the short-term and long-term bonds separately. Short-term debt's disciplining benefit not only yields a higher maximum revenue relative to long-term debt, but also allows the government to raise that maximum revenue at a lower level of debt issuance, which it will need to pay off next period.

Although the short-term bond is more effective at raising revenue than the long-term bond, it

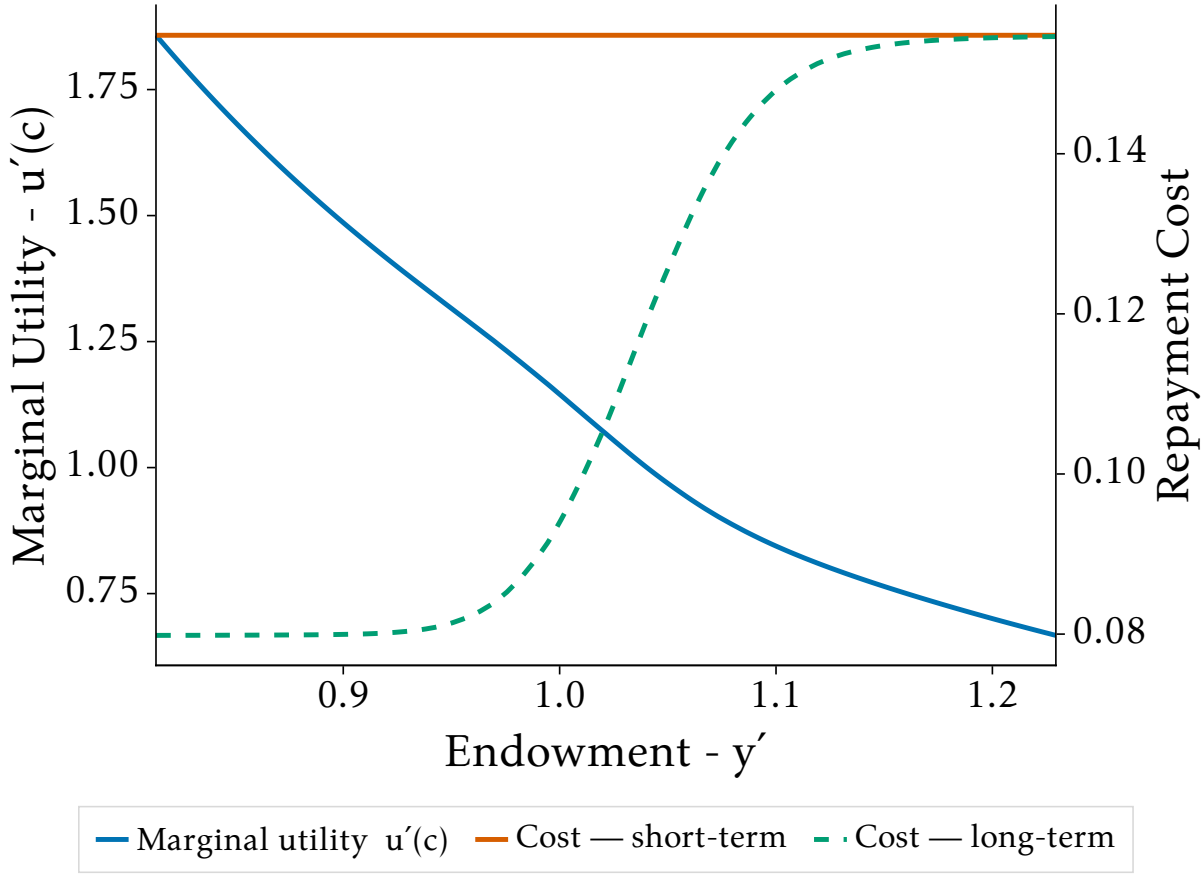


Figure 7: Hedging Property of Long-Term Debt

does not possess any hedging properties. This can be seen by comparing the right-hand sides of (15) and (17). Setting $\mathcal{E}_t = \mathcal{E}_{t+1} = 1$, the real cost of repaying the short-term bond is constant and invariant to the next period's income realization.

In contrast, the real cost of repaying one unit of the long-term bond is given by $1 + (1 - \delta)q^{LF}_{t+1}$. Note that bond prices are procyclical: as income falls, default risk rises and bond prices fall, meaning that the market value of unmatured long-term debt falls while marginal utility of consumption rises. This intuition is visually conveyed in Figure 7. Conditional on repayment, I use the government's equilibrium policy function for consumption and equilibrium pricing function for long-term debt to plot how marginal utility of consumption and the real cost of repaying the same amount of short-term and long-term debt vary with respect to income.

Figure 7 shows that repaying the long-term bond becomes cheaper precisely in states of the world where marginal utility of consumption is high, while the real cost of repaying the short-term bond remains invariant to the marginal utility of consumption. Note that if income were low enough, the real cost of repaying the long-term bond would flatten like the short-term bond.

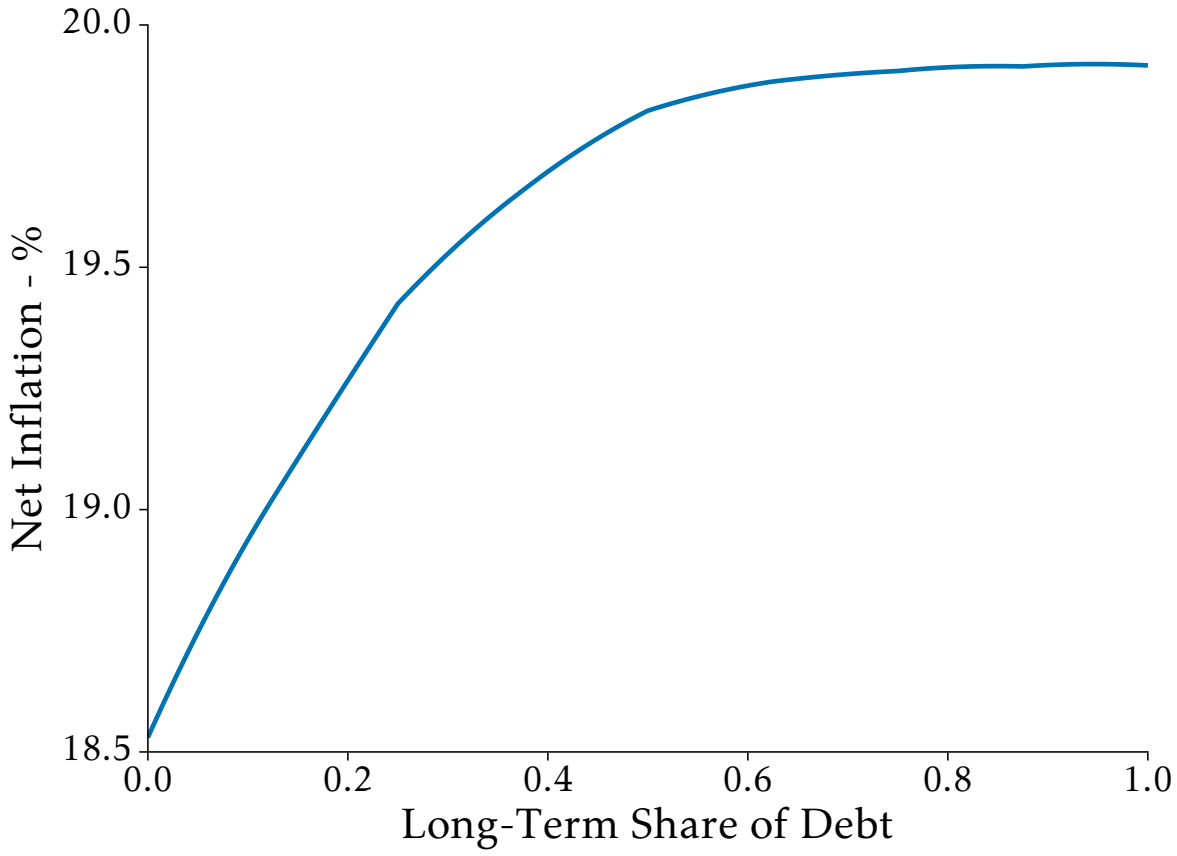


Figure 8: Maturity's Effects on Optimal Inflation

However, in those low-income states, the government would default in equilibrium.

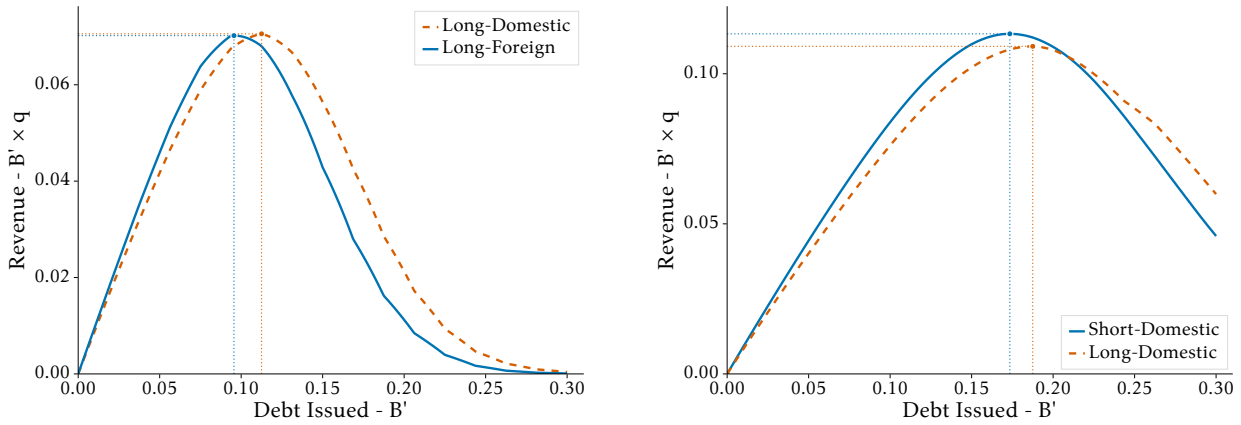
IV.D Both Margins

Lastly, I compare the short-term foreign currency bond against the long-term domestic currency bond.

I begin by showing that the long-term domestic currency bond has the worst incentive costs of all four bonds. In the full model, the government's intratemporal optimality condition for inflation is:

$$u_C(C_t)\left[\frac{1}{\Pi_t^2}\right]\left[b_t^{LD} + b_t^{SD} + q^{LD}(1-\delta)b_t^{LD}\right] = l_\Pi(\Pi_t - \Pi^*). \tag{20}$$

As before, optimal inflation is increasing in the amount of domestic currency debt owed, but the inclusion of long-term domestic currency debt adds a term to (20), $q^{LD}(1-\delta)b_t^{LD}$, which is the market value of the unmatured long-term domestic currency debt. When there is long-term domestic currency debt owed, the government becomes tempted to use inflation to erode not only present domestic currency debt obligations, but also the unmatured share of the long-term



(a) Long-Domestic vs. Long-Foreign

(b) Long-Domestic vs. Short-Domestic

Figure 9: Long-Domestic Revenue Curves

domestic currency bond that is due in future periods.

Thus, holding the amount of domestic currency debt constant, optimal inflation is increasing in the share of that debt that is long-term. In other words, one unit of long-term domestic currency debt increases the government's temptation to inflate more than one unit of short-term domestic currency debt, with this effect being stronger the longer the maturity of the long-term domestic currency bond. I show the effect of this additional channel in Figure 8, where I plot the optimal net inflation while only varying the share of domestic currency debt that is long-term along the x-axis.

Issuing one unit of the long-term domestic currency bond increases lenders' expectations of future inflation more than an equivalent unit of short-term domestic currency debt. Combined with the dilution risk present in all long-term debt, as shown in Section IV.C, this makes the long-term domestic currency bond the most expensive to issue.

Figure 9 compares the revenue curves of the long-term domestic currency bond against the long-term foreign currency bond and the short-term domestic currency bond. Figure 4 and Figure 6 already showed that both of these bonds have higher incentive costs than the short-term foreign currency bond.

The long-term domestic currency bond is unambiguously worse than the short-term domestic currency bond in terms of revenue, as the government can simultaneously raise more revenue while also having to borrow less in the future by issuing the short-term domestic bond instead. When compared against the long-term foreign currency bond, the long-term domestic currency bond's Laffer curve peaks at a slightly higher level, but a much higher level of issuance is required to reach that peak.[5]

On the other hand, the individual insurance benefits of domestic currency and long-term debt

[5]If the target inflation rate were sufficiently high, then the long-term domestic currency bond's Laffer curve peak would also be lower.

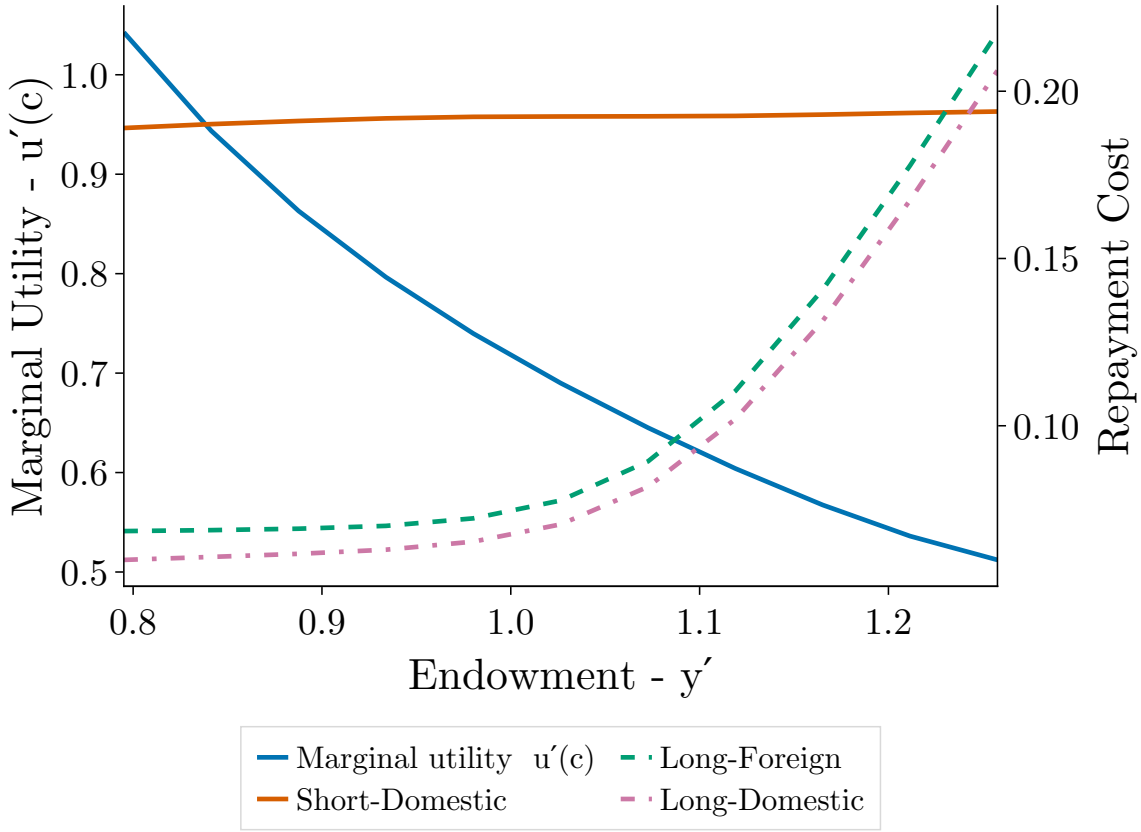


Figure 10: Long-Term Domestic Currency Debt's Hedging Properties

also compound to give the long-term domestic currency bond the most hedging benefits of all four bonds. As the right-hand side of (18) shows, in bad states of the world where the marginal utility of consumption is high, the real cost of repaying the long-term domestic currency bond decreases due to both inflation and the lower market value of the unmatured long-term domestic currency bond.

In Figure 10, I plot how the marginal utility of consumption and the real cost of repaying the same amount of (i) short-term domestic currency debt, (ii) long-term foreign currency debt, and (iii) long-term domestic currency debt vary with respect to income in equilibrium, holding the exchange rate fixed at unity.

As in Figure 7, the real cost of repayment of all three bonds is procyclical. Repaying all three bonds becomes cheaper precisely in states of the world where marginal utility of consumption is high. However, the cumulative hedging properties of the long-term domestic currency bond from both the dilution channel and inflation channel result in its real cost of repayment covarying the most negatively with marginal utility of consumption. In other words, holding the amount of debt owed constant, the long-term domestic currency bond offers the sovereign the most flexibility to smooth consumption.

IV.E Summary

Ultimately, this section first shows the disciplining-hedging trade-off when two bonds vary along either the currency or maturity dimension in isolation. Bonds denominated in foreign currency or with short maturity offer disciplining benefits to the government, while bonds denominated in domestic currency or with long maturity possess hedging properties against income and exchange rate risk.

I then show that when viewed through the lens of the disciplining-hedging trade-off, the maturity and currency aspects of each bond interact by compounding their respective disciplining and hedging properties when studied jointly. Out of the four bonds, the short-term foreign currency bond has the most disciplining benefits but no hedging benefits, and vice versa for the long-term domestic currency bond. Meanwhile, the short-term domestic currency bond and long-term foreign currency bond are intermediate assets that possess some disciplining and hedging properties, but not as much as the other two extreme bonds.

V Quantitative Analysis

In this section, I calibrate the model to match salient features of the Turkish economy. I choose Türkiye as my case study for two main reasons. First, Türkiye issues substantial amounts of debt in both domestic and foreign currency (as shown in Table 10), and it is covered by daily local-currency spread data, which I use to discipline the calibration. Second, the Turkish lira has experienced a series of significant depreciations since 2018, which provides a natural environment for assessing the model's portfolio choice predictions in a period of economic stress.

I first calibrate the model to a period of relative economic stability in Türkiye and present the portfolio choice moments it generates. I then apply the model to the Turkish lira crisis that began in 2018 and compare the model's predictions against the observed changes in Türkiye's issuance portfolio during the crisis. Ultimately, this section shows that the conventional disciplining-hedging trade-off and the compounding effects of currency and maturity can rationalize how the Turkish government has issued debt during periods of both economic stability and distress.

V.A Calibration

To evaluate the model's quantitative predictions for the sovereign's dynamic portfolio choice, I solve the model numerically via value function iteration. The full quantitative model also has three additional features relative to the model presented in Section III.

First, I follow Chatterjee and Eyigungor (2012) by having the short-term and long-term bonds pay coupons, denoted by κ^S and κ^L, respectively. Furthermore, I follow Aguiar et al. (2016) by setting $\kappa^S = R^*$ and $\kappa^L = \delta + R^* - 1$. This has the convenient property of normalizing the risk-free price of both foreign currency bonds to 1, which facilitates comparison of bond prices across

different maturities.[6] The yields to maturity of short-term and long-term bonds are then given by $\frac{R^*}{q} - 1$ and $\frac{\delta + R^* - 1}{q} - \delta$, respectively.

Second, I subject the sovereign's discrete default-repayment decision to additive taste shocks that are i.i.d. and Type-1 Extreme-Value (Gumbel) distributed. As shown in Chatterjee and Eyigungor (2012), it is notoriously difficult to achieve numerical convergence in sovereign debt models with long-term debt without any randomization. I set the location parameter to $-\alpha(\gamma + \ln(2))$ where γ is the Euler-Mascheroni constant, and α is the scale parameter of the shocks. The choice of the location parameter ensures that the taste shocks have mean zero and thus do not give any ex ante utility to either default or repayment. This assumption replaces the sovereign's deterministic default-repayment decision with a probabilistic one, which smooths the bond pricing functions and yields a closed-form expression for the ex ante probability of repayment (before observing the taste-shock realization).

Finally, I assume that the household has recursive (Epstein-Zin) preferences as in Epstein and Zin (1991) and Weil (1990), which allows for the household's risk aversion and intertemporal elasticity of substitution to be separated. The household's lifetime utility V_t is given recursively by:

$$V_t = \left[(1-\beta) \left(C_t^{1-\frac{1}{\rho}} - \left(1 - \tfrac{1}{\rho}\right) l(\Pi_t - \Pi^*) \right) + \beta \left(\mathbb{E}_t \left[V_{t+1}^{1-\sigma} \right] \right)^{\frac{1-1/\rho}{1-\sigma}} \right]^{\frac{1}{1-1/\rho}}, \tag{21}$$

where σ is the coefficient of relative risk aversion and ρ is the intertemporal elasticity of substitution. When $\sigma = 1/\rho$, (21) is, up to a monotone transformation, the expected utility specification of Section III. Separating the two parameters is important because the household needs to be sufficiently risk-averse to avoid issues of numerical indifference. Epstein-Zin preferences then allow the household to be sufficiently risk-averse while not imposing a low intertemporal elasticity of substitution, which standard preferences with constant relative risk aversion (CRRA) utility would require.

Two further numerical challenges arise when solving the model. First, due to the curse of dimensionality, conventional discrete-choice grid-based methods are computationally infeasible.[7] Second, the state space over which the policy functions for bond issuance are well-defined is irregular. To overcome these challenges, I use Gaussian process regression to approximate all equilibrium functions in the model. I provide details of my numerical algorithm in Appendix C.

I discipline my model by calibrating it to the Turkish economy over the period 2004:I–2018:II. My sample period begins in 2004:I because the Turkish lira only became free-floating after 2001, and Turkish inflation only became relatively stable from 2004 onwards. I end in 2018:II because that was when the lira crisis began.

One period in my model corresponds to one year. I impose the following functional form for

[6] The risk-free price of a foreign currency bond with decay parameter δ is given by $\frac{\kappa}{\delta + R^* - 1}$.

[7] This restriction precludes the use of taste shocks to the sovereign's bond issuance decision, which is why all my policy functions for bond issuance and inflation are deterministic.

the sovereign's period utility function, which is separable in consumption and inflation:

$$u(C_t) - l(\Pi_t - \Pi^*) = \frac{C_t^{1-1/\rho}}{1-1/\rho} - \frac{\psi}{2}(\Pi_t - \Pi^*)^2, \tag{22}$$

where ρ is the household's intertemporal elasticity of substitution, ψ is the scale of the disutility of inflation (so that $\ell(x) = x^2/2$), and Π^* is the inflation target. Following Chatterjee and Eyigungor (2012), I assume that the output costs of default are given by the following functional form:

$$h(\mathcal{Y}_t) = \mathcal{Y}_t - \max\left\{0, \lambda_0 \mathcal{Y}_t + \lambda_1 \mathcal{Y}_t^2\right\}. \tag{23}$$

Lastly, the two shocks in my model follow the process:

$$\begin{pmatrix} \log(\mathcal{Y}_t) \\ \log(\mathcal{E}_t) \end{pmatrix} = \begin{pmatrix} \rho_y & 0 \\ 0 & \rho_e \end{pmatrix} \begin{pmatrix} \log(\mathcal{Y}_{t-1}) \\ \log(\mathcal{E}_{t-1}) \end{pmatrix} + \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix} \quad \text{where} \quad \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix} \sim N\left(0, \begin{pmatrix} \sigma_y^2 & \rho_{y,e}\sigma_y\sigma_e \\ \rho_{y,e}\sigma_y\sigma_e & \sigma_e^2 \end{pmatrix}\right).$$

ρ_y and ρ_e are the persistence parameters of the endowment and real exchange rate, respectively, while σ_y and σ_e denote the volatility of the two shocks. Finally, $\rho_{y,e}$ is the correlation between the two shocks, which is typically negative for emerging market economies.

With these functional forms, 18 parameter values must be specified. I summarize parameter values that were selected directly in Table 5 and parameters that were selected by matching moments from the Turkish economy in Table 6.

Parameter	Description	Value	Source
σ	risk aversion	5	Ottonello and Perez (2019)
ρ	intertemporal elasticity of substitution	0.5	standard value
R^*	risk-free rate	1.04	1-year US Treasury yield
δ	perpetuity decay parameter	0.1	10-year maturity
κ^S	short-term coupon	1.04	Normalization
κ^L	long-term coupon	0.14	Normalization
θ	re-entry probability	0.17	6-year exclusion
ρ_y	output persistence	0.728	data
σ_y	output volatility	0.028	data
ρ_e	real exchange rate persistence	0.879	data
σ_e	real exchange rate volatility	0.096	data
$\rho_{y,e}$	shock correlation	−0.326	data
α	taste shock scale	0.01	numerical convergence

Table 5: Parameters Selected Directly

I collect data from FRED on Türkiye's real GDP, nominal exchange rate with the US, CPI, and US CPI, where the latter three series are used to compute the real exchange rate. I linearly detrend the data and estimate the parameters of the joint endowment–real exchange rate process

Parameter	Description	Value
β	discount factor	0.86
λ_0	default cost: linear	−0.481
λ_1	default cost: quadratic	0.516
ψ	inflation disutility	1.1
Π^*	inflation target	1.083

Table 6: Parameters Selected by Matching Moments

via maximum likelihood estimation. Due to the shortness of the empirical sample, I estimate the parameters using quarterly-frequency data and then annualize them.[8] The estimated parameters are similar to the annual-frequency values used in Ottonello and Perez (2019), and the persistence of the output process is also close to the values estimated in Arellano (2008) and Chatterjee and Eyigungor (2012) once annualized.

I set the household's risk aversion and intertemporal elasticity of substitution to $\sigma = 5$ and $\rho = 0.5$, respectively. The real business cycle literature, which often uses expected utility preferences, typically sets the CRRA coefficient to 2, which corresponds to an intertemporal elasticity of substitution of 0.5. I set the scale parameter of the Gumbel-distributed taste shock to $\alpha = 0.01$, which is small enough to ensure that the sovereign's default-repayment decision is not dominated by taste shocks, but large enough for numerical convergence.

The parameters describing the maturity of the bonds were set as follows. Over 2004:I–2018:II, the average maturity of the foreign currency debt that Türkiye issued was 8.6 years.[9] Because the average maturity of domestic and foreign currency debt that my model generates will be a convex combination of the 1-year and long-term bonds, I set the decay parameter of the long-term bond to correspond to 10 years in maturity, or $\delta = 0.1$.[10] I use the same decay parameter for the long-term domestic and foreign currency bonds. Choosing a shorter one for the domestic bond to match its observed maturity would build the maturity gap into the calibration by assumption.

I follow Arellano and Ramanarayanan (2012) and set the re-entry probability to $\theta = 0.17$, which corresponds to an average exclusion period of 6 years. Finally, the gross risk-free rate was set to $R^* = 1.04$, which is roughly the annualized yield of a 1-year US Treasury bond.

The remaining five parameters, β, λ_0, λ_1, ψ, and Π^*, are selected by matching (i) an average external debt-to-GDP ratio of 12.6%, (ii) an average spread of 10.3% for 1-year bonds issued by the Turkish government, (iii) a standard deviation of 3.0% for spreads, (iv) an average annualized inflation rate of 8.6%, and (v) a standard deviation of inflation of 1.8%. Data on inflation and the external trade balance were computed from FRED and the World Bank's World Development

[8] I use a Cholesky decomposition to transform the correlated bivariate normal variables into two independent normal variables. I then use Gauss-Hermite quadrature with 121 nodes (11 nodes per shock) as a tensor product to compute expected values.

[9] The average maturity of Turkish lira-denominated debt over this period was 3.1 years.

[10] Data on domestic currency spreads is also only available for bonds of 1, 3, 5, 7, and 10 years in maturity.

Statistic	Data	Model
Mean debt-to-GDP ratio	12.6%	10.9%
Mean spread (1 year, domestic currency)	10.3%	9.6%
Std. spread (1 year, domestic currency)	3.0%	2.1%
Mean inflation	8.6%	8.7%
Std. inflation	1.8%	1.4%

Table 7: Model Results and Data Moments

Indicators, respectively.

In order to compute empirical moments on spreads, I use data from Du and Schreger (2016), Du and Schreger (2022), and Du et al. (2025), which provide daily-frequency data on the difference between the yield of a Turkish lira-denominated bond and a US Treasury bond of the same tenor. For the spread between USD-denominated debt issued by the Turkish government and US Treasury bonds of the same maturity, I use data from Broner et al. (2013). However, their sample for Türkiye only runs from 1996:II to 2009:II.

Although all five parameters are jointly determined, each is nonetheless most closely tied to one empirical moment, which offers a heuristic sense of identification. The discount factor, β, is most closely tied to the average debt-to-GDP ratio because it shapes the household's incentive to front-load consumption. As discussed in Chatterjee and Eyigungor (2012), the parameters λ_0 and λ_1 mostly affect the average and standard deviation of spreads, respectively, because they determine how costly default is for the sovereign, and hence how attractive default is. Because data on foreign currency spreads end in 2009:II, I instead use the spread between 1-year domestic currency bonds and 1-year US Treasury bonds. While this differential encompasses both default risk and inflation risk, Du and Schreger (2016) show that the default risk premium is the dominant component of spreads for emerging market economies.

Lastly, the inflation target, Π^*, and the disutility parameter, ψ, affect the average and standard deviation of inflation, respectively. The inflation target, Π^*, is set to 1.083, or 8.3%. This is slightly lower than the average inflation rate of 8.6%, but as long as the government owes any domestic currency debt of either maturity, it will choose an inflation level slightly higher than its target.

V.B Simulation Results

I now present the model's quantitative predictions for the currency and maturity composition of the debt that the sovereign issues. The domestic currency share is the share of total debt issued that is denominated in domestic currency:

$$\frac{q_t^{SD} b_{t+1}^{SD} + q_t^{LD}(b_{t+1}^{LD} - (1-\delta)b_t^{LD}/\Pi_t)}{q_t^{SD} b_{t+1}^{SD} + q_t^{LD}(b_{t+1}^{LD} - (1-\delta)b_t^{LD}/\Pi_t) + \mathcal{E}_t(q_t^{SF} b_{t+1}^{SF} + q_t^{LF}(b_{t+1}^{LF} - (1-\delta)b_t^{LF}))}. \tag{24}$$

Statistic	Data	Model
Domestic currency share	89.7%	81.7%
Mean maturity (domestic currency)	3.1 years	2.2 years
Mean maturity (foreign currency)	8.6 years	5.8 years
Mean spread (1 year, foreign currency)	1.49%	1.23%
Mean spread (10 year, foreign currency)	4.7%	3.3%
Mean spread (10 year, domestic currency)	7.2%	9.9%

Table 8: Untargeted Moments

For a given currency denomination, the average maturity of debt in the model is the weighted average of the bonds' maturities, with weights equal to each bond's share of the debt issued in that currency. Short-term bonds have a maturity of 1 year and long-term bonds have a maturity of $1/\delta = 10$ years. Specifically, for foreign currency debt, the weight for short-term debt is:

$$\frac{q_t^{SF} b_{t+1}^{SF}}{q_t^{SF} b_{t+1}^{SF} + q_t^{LF}(b_{t+1}^{LF} - (1-\delta) b_t^{LF})}, \tag{25}$$

and for domestic currency debt, the weight for short-term debt is:

$$\frac{q_t^{SD} b_{t+1}^{SD}}{q_t^{SD} b_{t+1}^{SD} + q_t^{LD}(b_{t+1}^{LD} - (1-\delta) b_t^{LD}/\Pi_t)}. \tag{26}$$

Table 8 shows that the model is able to match key features of the Turkish economy even though these moments are not targeted. First, it predicts a domestic currency share of 81.7%, compared with 89.7% in the data. Second, it generates the maturity gap between domestic and foreign currency debt, which I discuss below.

The model predicts that the average maturity of issued domestic and foreign currency debt is 2.2 years and 5.8 years, respectively. Both of these values are lower than their empirical counterparts, but this is likely because the only two maturities available for the sovereign to choose among are 1 and 10 years, whereas the Turkish government has a continuum of maturity choices available to it. This difficulty is common to quantitative sovereign default models with maturity choice: Bocola and Dovis (2019) and Mihalache (2020) impose a disutility from deviating from a target maturity to match the average maturity of debt in their respective economies.

Nonetheless, the model is able to endogenously generate the maturity gap between issued domestic and foreign currency debt despite not being targeted to do so. In the model, foreign currency debt has 2.6 times the average maturity of domestic currency debt, compared with 2.8 times in the data.

To understand the model's predictions regarding the state-dependent nature of the government's borrowing decisions, I present a heatmap of the government's portfolio policy functions across the exogenous state space in Figure 11. Specifically, Figure 11 shows the amount-weighted

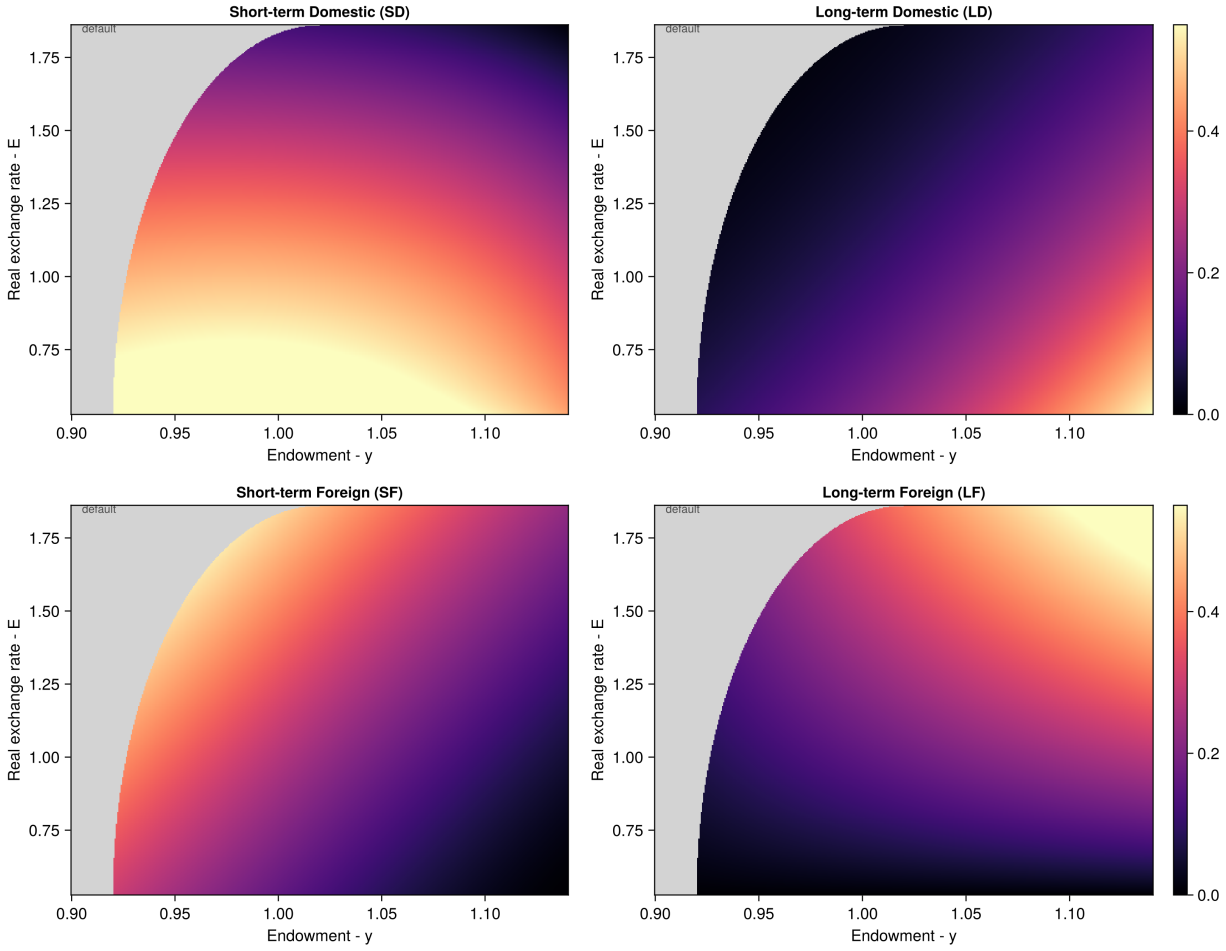


Figure 11: Heatmap of Portfolio Policy Functions

share of each bond's issuance in the government's portfolio across different realizations of the exogenous state variables, $\mathcal{Y}_t$ and $\mathcal{E}_t$.

I evaluate the policy functions at the average debt portfolio $\mathbb{B}_t$ from a 100,000-period simulation of the model, discarding the first 5,000 periods and keeping only periods in which the sovereign has access to financial markets.[11]

On the left-hand side of each panel in Figure 11, income is so low that the government would default in equilibrium, whatever the real exchange rate. As the real exchange rate rises along the y-axis, raising the cost of repaying foreign currency debt, a higher realization of $\mathcal{Y}_t$ is required for

[11] Varying $\mathbb{B}_t$ primarily affects the heatmap by altering the set of states in which the government would default in equilibrium. Higher $\mathbb{B}_t$ results in a larger set of states in which the government would default, and the amount of foreign currency debt owed shifts the gradient of the default set with respect to the real exchange rate realization. Nonetheless, the qualitative conclusion that the short-term foreign currency bond and the long-term domestic currency bond only start dominating the government's issuance portfolio during economic downturns and upturns, respectively, is robust to the choice of $\mathbb{B}_t$.

the government to prefer repayment in equilibrium.

The top-left and bottom-right quadrants of each panel in Figure 11 together show that the extreme trade-offs involved in issuing short-term foreign currency debt and long-term domestic currency debt result in these bonds being issued only in extreme states of the economy. The short-term foreign currency bond dominates the issuance portfolio only close to the default frontier, where the government is simultaneously experiencing a significant recession and depreciation of its domestic currency. In such states, where marginal utility of consumption is highest, the government's primary motive is to raise as much revenue as possible. Moreover, due to mean reversion, the government is likely to enjoy a more favorable state of the world tomorrow, where the lack of hedging from the short-term foreign currency bond is less costly in expectation.

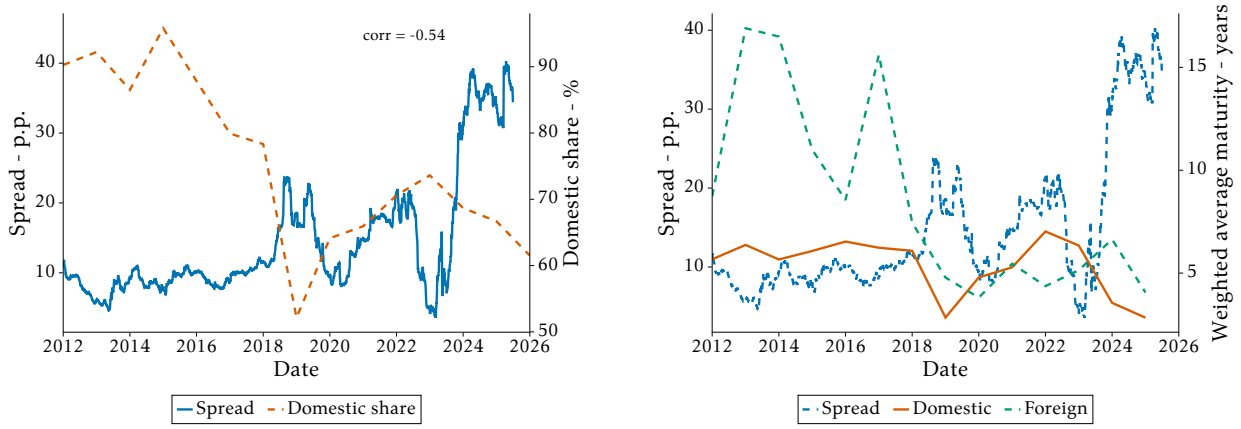
In such states, the long-term domestic currency bond does not get issued at all because its outsized incentive costs outweigh its hedging benefits. Conversely, the long-term domestic currency bond's share of the issuance portfolio peaks, at roughly one half, when the government is jointly enjoying an economic boom and a significant appreciation of its domestic currency. In those appealing states, marginal utility of consumption is low, so the government is less concerned with maximizing current revenue, which makes the long-term domestic currency bond's incentive costs less binding.

Here, the government's primary motive is to hedge the household against future low income and depreciation of the domestic currency, which raise the real cost of foreign currency debt precisely when marginal utility is high. A portfolio concentrated in short-term foreign currency debt would offer no state contingency in those bad states, so the short-term foreign currency bond is not issued at all. Both forces make the long-term domestic currency bond appealing only in these states.

V.C Lira Crisis

In late 2018, the Turkish lira began a series of significant depreciations against the US dollar, from approximately 3.64 lira per US dollar in 2017 to 39.49 by the end of 2025. The Turkish economy also experienced a sharp quarterly contraction of 9.9% in 2020:II during the COVID-19 pandemic, as well as a period of unorthodox monetary policy that raised inflation from 11.92% in 2017 to 30.89% in 2025.

Figure 12 shows the time series of Türkiye's issuance portfolio and 1-year spread in the data. Specifically, Figure 12a shows the time series of the share of debt denominated in Turkish lira for a given year, while Figure 12b shows the time series of the weighted average maturity of domestic currency debt and of foreign currency debt, separately. Consistent with the mechanisms of the model, the deterioration in the Turkish economy during this period is associated with sharp increases in the 1-year spread as investors demanded higher compensation for the increased risk. Moreover, the data on Türkiye's issuance portfolio shows that as Türkiye's economy deteriorated, the government issued a larger share of its debt in foreign currency and shortened the maturity of issued debt, regardless of currency denomination.



(a) Currency Share

(b) Weighted Average Maturity

Figure 12: Time Series of Türkiye's Issuance Portfolio and 1-Year Spread

	Data	**Model**
Δ Mean Inflation	+25.42 p.p.	+19.36 p.p.
Δ Domestic Currency Share	−20.80 p.p.	−26.73 p.p.
Δ Mean Maturity (Domestic Currency)	−1.03 years	−1.21 years
Δ Mean Maturity (Foreign Currency)	−8.87 years	−2.03 years

Δ: post-crisis (2018–2025) minus pre-crisis (2012–2017); p.p.: percentage points.

Table 9: Model Validation: Türkiye's Lira Crisis

I therefore conduct an external validation exercise to test whether the model replicates these observed changes in Türkiye's issuance portfolio during this period of economic stress. Specifically, I compare the model's predictions regarding the change in the domestic currency share and average maturity of debt against the data. For 2012–2017, I feed the observed detrended real GDP and real exchange rate into the model and record its predictions for the domestic currency share, inflation, and average maturity of domestic and foreign currency debt.

To help the model reproduce the observed surge in Türkiye's inflation during this period, I reduce the inflation disutility parameter ψ from 1.1 to 0.05 as a reduced-form way to model the unorthodox monetary policy that was implemented during this period. I then re-solve the model and record its predictions for 2018–2025.

The results of this exercise are shown in Table 9. The reduction in ψ allows the model to generate a sizable inflation surge (19.4 percentage points, compared with 25.4 in the data). In principle, further reducing ψ and/or increasing the government's target inflation rate would allow for an even larger inflation surge. However, changes in the government's preferences over inflation would be factored into the equilibrium pricing functions for all domestic currency debt,

thereby making domestic currency debt too expensive to issue in equilibrium. Not only would this compromise the model's ability to match the observed shift of domestic currency debt, but it may also reduce average inflation. While inflation would always be higher in the initial periods because there is still domestic currency debt owed, the reduction or outright elimination of domestic currency debt issued and thus owed in later periods would reduce the government's incentive to inflate.

The model performs reasonably well in matching the observed changes in Türkiye's issuance portfolio during the crisis, as the disciplining benefits of short-term and foreign currency debt become more valuable. The model predicts a 26.73-percentage-point decrease in the share of debt denominated in domestic currency and a 1.21-year decrease in the average maturity of domestic currency debt, which are both close to their empirical counterparts. Notably, the model predicts only a 2.03-year decrease in the average maturity of foreign currency debt, compared with 8.87 years in the data. This is likely because maturity is modeled as a discrete choice between 1 and 10 years. This creates two challenges for the model's ability to match the data in terms of maturity. First, the average maturity of Türkiye's foreign currency debt over 2012–2017 was 12.88 years, whereas the highest maturity the model is able to generate is 10 years. Second, fixing the short-term bond's maturity at 1 year, rather than at a more intermediate maturity such as 2 or 3 years, may also be limiting. This issue may be ameliorated if maturity were modeled as a continuous choice, as in Bocola and Dovis (2019). However, this would require a more complex model solution method and is beyond the scope of this paper.

Nonetheless, the results of this exercise show that the model is able to qualitatively match the observed changes in Türkiye's issuance portfolio during the crisis. Together, the results demonstrate that the conventional disciplining-hedging trade-off and the compounding effects of currency and maturity can rationalize how the Turkish government has issued debt both in normal times and in extreme states.

VI Conclusion

In this paper, I study how emerging market economies jointly choose the currency and maturity composition of their debt. Issuance data from a wide range of emerging market economies over a long time span shows that sovereigns' domestic currency debt is often of significantly shorter maturity than the foreign currency debt that they issue. Additionally, periods of higher sovereign spreads are associated with sovereigns issuing a higher share of their debt in foreign currency and at shorter maturities.

To understand the mechanisms behind these patterns, I build a dynamic model of sovereign debt with endogenous default and inflation. The government solves a portfolio choice problem with four bonds: (i) short-term, domestic currency bonds, (ii) long-term, domestic currency bonds, (iii) short-term, foreign currency bonds, and (iv) long-term, foreign currency bonds. The government's lack of commitment to repay its debt, refrain from inflation, and avoid over-borrowing in

the future, together with income and real exchange rate dynamics, generates a set of trade-offs between the four bonds that explains the empirical patterns observed in the data.

In particular, I show that portfolio choice is state dependent. In normal times, governments rely on the two intermediate bonds, short-term domestic currency debt and long-term foreign currency debt, because the two extreme bonds are too costly relative to their benefits. Domestic currency debt is therefore rarely of longer maturity than foreign currency debt: long-term domestic currency debt's high hedging benefits are outweighed by its high incentive costs except in economic booms. Likewise, foreign currency debt is seldom short-term, because its strong disciplining benefits are only valuable to the government in times of extreme economic distress. Calibrated to Türkiye, the model generates a foreign-to-domestic maturity ratio of 2.6, close to the 2.8 in the data, and qualitatively reproduces the shift away from domestic currency debt and toward shorter maturities during the lira crisis.

Several extensions remain. Allowing foreign lenders to be risk-averse rather than risk-neutral is a natural one, although I expect it to leave the qualitative results unchanged, and if anything to strengthen them: the additional risk premium would amplify the incentive costs of long-term or domestic currency debt. Another is to endogenize the maturity of the long-term bonds, as in Bocola and Dovis (2019), rather than fixing it at 10 years.

Taken together, the evidence and the calibrated model show that sovereign debt portfolios are state dependent. In normal times, governments rely on short-term domestic currency and long-term foreign currency debt, and the extreme instruments appear only at the edges of the state space. The maturity gap between domestic and foreign currency debt is the natural outcome of governments balancing disciplining against hedging on two margins at once.

References

Aguiar, M., M. Amador, E. Farhi, and G. Gopinath (2013): "Crisis and Commitment: Inflation Credibility and the Vulnerability to Sovereign Debt Crises," Tech. Rep. w19516, National Bureau of Economic Research, Cambridge, MA.

Aguiar, M., S. Chatterjee, H. Cole, and Z. R. Stangebye (2016): "Quantitative Models of Sovereign Debt Crises," in *Handbook of Macroeconomics*, Elsevier, vol. 2, 1697–1755.

Arellano, C. (2008): "Default Risk and Income Fluctuations in Emerging Economies," *American Economic Review*, 98, 690–712.

Arellano, C. and A. Ramanarayanan (2012): "Default and the Maturity Structure in Sovereign Bonds," *Journal of Political Economy*, 120, 187–232.

Arslanalp, S. and T. Tsuda (2012): "Tracking Global Demand for Advanced Economy Sovereign Debt," *IMF Working Papers*, 12, 1.

Bocola, L. and A. Dovis (2019): "Self-Fulfilling Debt Crises: A Quantitative Analysis," *American Economic Review*, 109, 4343–4377.

Broner, F. A., G. Lorenzoni, and S. L. Schmukler (2013): "Why Do Emerging Economies Borrow Short Term?" *Journal of the European Economic Association*, 11, 67–100.

Chatterjee, S. and B. Eyigungor (2012): "Maturity, Indebtedness, and Default Risk," *American Economic Review*, 102, 2674–2699.

Du, W., R. Keerati, and J. Schreger (2025): "Decoupling Dollar and Treasury Privilege," *International Finance Discussion Paper*, 1.

Du, W., C. E. Pflueger, and J. Schreger (2020): "Sovereign Debt Portfolios, Bond Risks, and the Credibility of Monetary Policy," *The Journal of Finance*, 75, 3097–3138.

Du, W. and J. Schreger (2016): "Local Currency Sovereign Risk," *The Journal of Finance*, 71, 1027–1070.

——— (2022): "Sovereign Risk, Currency Risk, and Corporate Balance Sheets," *The Review of Financial Studies*, 35, 4587–4629.

Engel, C. and J. Park (2022): "Debauchery and Original Sin: The Currency Composition of Sovereign Debt," *Journal of the European Economic Association*, 20, 1095–1144.

Epstein, L. G. and S. E. Zin (1991): "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: An Empirical Analysis," *Journal of Political Economy*, 99, 263–286.

Gu, G. W. and Z. R. Stangebye (2023): "Costly Information and Sovereign Risk," *International Economic Review*, 64, 1397–1429.

Gürkaynak, R. S., B. Sack, and J. H. Wright (2007): "The U.S. Treasury Yield Curve: 1961 to the Present," *Journal of Monetary Economics*, 54, 2291–2304.

Hatchondo, J. C. and L. Martinez (2009): "Long-Duration Bonds and Sovereign Defaults," *Journal of International Economics*, 79, 117–125.

Hatchondo, J. C., L. Martinez, and H. Sapriza (2010): "Quantitative Properties of Sovereign Default Models: Solution Methods Matter," *Review of Economic Dynamics*, 13, 919–933.

Hatchondo, J. C., L. Martinez, and C. Sosa-Padilla (2016): "Debt Dilution and Sovereign Default Risk," *Journal of Political Economy*, 124, 1383–1422.

Lee, A. S. (2021): "Why Do Emerging Economies Borrow in Foreign Currency? The Role of Exchange Rate Risk," *SSRN Electronic Journal*.

Maeng, F. S. (2025): "Default, Inflation Expectations, and the Currency Denomination of Sovereign Bonds," Tech. rep., Nanyang Technological University.

Mihalache, G. (2020): "Sovereign Default Resolution through Maturity Extension," *Journal of International Economics*, 125, 103326.

Ottonello, P. and D. J. Perez (2019): "The Currency Composition of Sovereign Debt," *American Economic Journal: Macroeconomics*, 11, 174–208.

Schwert, M. (2017): "Municipal Bond Liquidity and Default Risk," *The Journal of Finance*, 72, 1683–1722.

Sunder-Plassmann, L. (2020): "Inflation, Default and Sovereign Debt: The Role of Denomination and Ownership," *Journal of International Economics*, 127, 103393.

Weil, P. (1990): "Nonexpected Utility in Macroeconomics," *The Quarterly Journal of Economics*, 105, 29.

A Data

A.A Data on Issuance

My individual bond-level data from Bloomberg spans 19 emerging market economies from 1990 until 2024.[12] I include countries in my sample if (i) Bloomberg has at least 200 valid bond issuances for that country over the sample period, and (ii) at least 5% of the country's total amount of issued debt is denominated in a foreign currency. I impose a minimum foreign currency threshold since the debt management offices of countries with very low foreign currency debt shares are unlikely to need to optimize their borrowing portfolio with respect to currency composition on top of maturity composition.[13]

There are two drawbacks of Bloomberg data. First, it is impossible to know whether the bond was bought by a domestic or international investor since there is no data on the creditor's nationality. To address this, I follow Gürkaynak et al. (2007) and Ottonello and Perez (2019) by removing all bonds issued by a country's central bank and bonds that mature in less than 3 months from issuance since they are typically only bought by domestic investors. Second, Bloomberg data also includes sub-national (or municipal) bonds. Since sub-national bonds have different risk profiles from national bonds,[14] I also remove them.[15]

A.B Data on Spreads

I use panel data on sovereign bond spreads from Datastream and Du and Schreger (2016). For a given country, both datasets provide a time series of the yield differential between a bond issued by that country and denominated in its domestic currency, and the yield on a bond of the same maturity but issued by the US Treasury and denominated in USD. The resulting yield differential is the sovereign bond spread, which is a measure of the risk premium that investors demand to hold a country's debt relative to a risk-free benchmark.

The data from Du and Schreger (2016) is of daily frequency and covers eight emerging market economies that overlap with my issuance data. In order to expand the coverage across countries, I further use data from Datastream. Unfortunately, Datastream data is only available at monthly, quarterly, or annual frequency, and the monthly frequency data contains gaps in the time series. Therefore, I use Du and Schreger (2016)'s data where possible, and use quarterly frequency data from Datastream for the remaining four countries.

[12] Bloomberg data only became available for most countries from 1990 onwards.

[13] This restriction excludes notable emerging market economies such as Brazil, China, India, Malaysia, and Lithuania. I omit Lithuania because it officially adopted the euro as its national currency in 2015 and has since denominated all its debt in euros. By amount issued, the foreign currency share of Brazil, China, India, and Malaysia's sovereign debt was only 2.9%, 1.50%, 0.19%, and 0.8%, respectively. Furthermore, India only started issuing non-rupee-denominated sovereign bonds in June 2019.

[14] Schwert (2017) finds that municipal bonds pay an oversized risk premium for default risk relative to comparable bonds at the national level.

[15] I filter out all municipal bonds, bonds issued to finance local projects, and bonds issued by state-owned companies. A bond is considered to be from a state-owned company if the issuer name contains strings such as 'Ltd.' or 'Corp.'

Country	Sample Period	Number of Bonds	FC Share	Weighted Average Maturity
Angola	Nov. 2005 – Dec. 2024	2,344	9.8%	4.1 years
Argentina	Mar. 1991 – Dec. 2024	874	62.1%	10.9 years
Colombia	Dec. 1990 – Dec. 2024	763	11.8%	11.7 years
Egypt	Dec. 1993 – Dec. 2024	2,715	9.5%	2.0 years
Hungary	Jul. 1993 – Dec. 2024	2,246	13.8%	5.5 years
Indonesia	Jul. 1990 – Dec. 2024	1,146	14.5%	10.6 years
Jamaica	Sep. 1991 – Dec. 2024	778	55.5%	7.3 years
Jordan	Aug. 1993 – Nov. 2024	1,229	17.8%	4.3 years
Mexico	Mar. 1990 – Dec. 2024	2,450	7.2%	4.9 years
Philippines	May 1990 – Dec. 2024	2,066	12.5%	6.5 years
Poland	Dec. 1993 – Nov. 2024	1,242	13.7%	6.4 years
Qatar	Feb. 1999 – Sep. 2024	238	55.0%	10.6 years
Romania	Jun. 1997 – Dec. 2024	888	38.0%	6.9 years
Russia	May 1993 – May 2024	571	12.4%	8.1 years
Tunisia	Sep. 1990 – Dec. 2024	281	21.7%	6.5 years
Türkiye	Mar. 1990 – Dec. 2024	992	15.8%	4.1 years
Ukraine	Mar. 1995 – Dec. 2022	1,813	42.8%	6.1 years
Uruguay	Jan. 1990 – Dec. 2024	348	52.2%	14.9 years
Venezuela	Dec. 1990 – Nov. 2018	1,926	35.0%	7.7 years

Table 10: Country-Level Issuance Data Description

Country	Sample Period	Number of Observations	Source
Colombia	Aug. 2003 – Dec. 2024	5,381	Du and Schreger (2016)
Egypt	Jan. 1997 – Oct. 2024	112	Datastream
Hungary	Jan. 2000 – Dec. 2024	6,450	Du and Schreger (2016)
Indonesia	Feb. 2003 – Dec. 2024	5,488	Du and Schreger (2016)
Jamaica	Jan. 1989 – Oct. 2024	144	Datastream
Mexico	Sep. 2002 – Dec. 2024	5,724	Du and Schreger (2016)
Philippines	Dec. 2000 – Dec. 2024	5,901	Du and Schreger (2016)
Poland	Jan. 2000 – Dec. 2024	6,452	Du and Schreger (2016)
Romania	Jan. 1994 – Oct. 2024	116	Datastream
Russia	Apr. 2006 – Feb. 2022	4,136	Du and Schreger (2016)
Türkiye	Feb. 2005 – Dec. 2024	5,107	Du and Schreger (2016)
Uruguay	Oct. 1991 – Oct. 2024	116	Datastream

Table 11: Country-Level Bond Spread Data Description

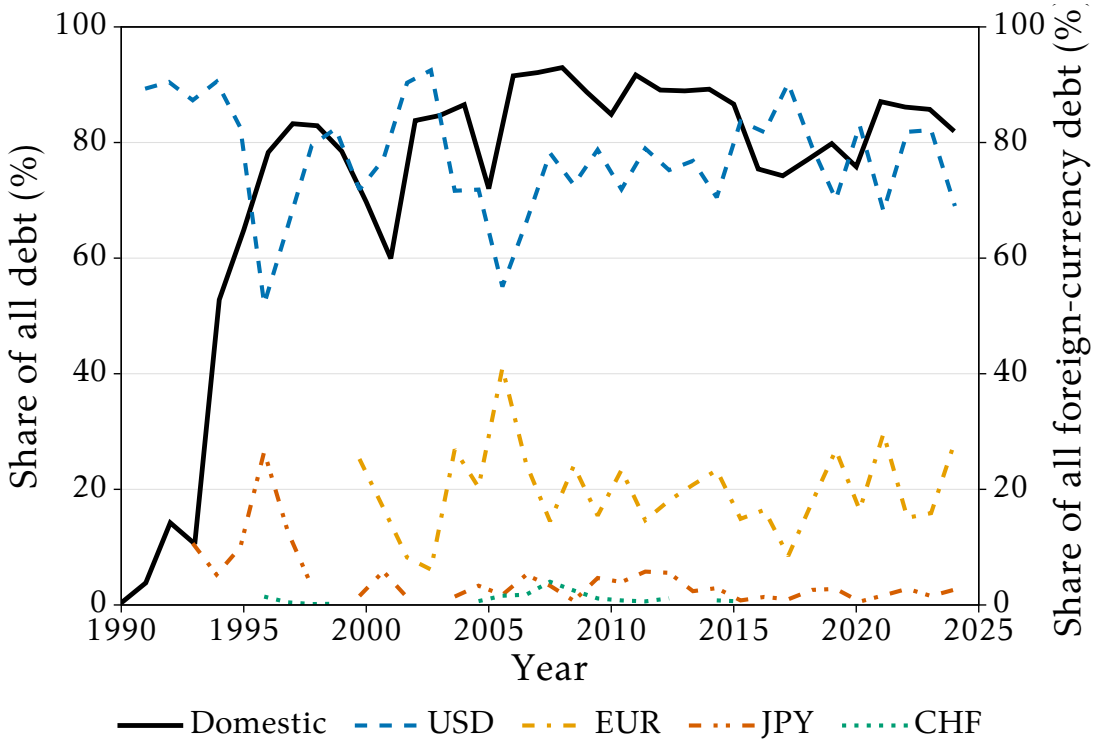


Figure 13: Time Series of the Currency Composition of Sovereign Debt

B Additional Empirical Results

	(1) Pooled	(2) Fixed Effects
Domestic currency	−7.802***	−6.862***
	(1.052)	(1.139)
Constant	12.497***	
	(1.056)	
Country fixed effects	No	Yes
Year fixed effects	No	Yes
Observations	24,909	24,909
R^2	0.145	0.296

Notes: The dependent variable is the maturity at issuance in years. *Domestic currency* is an indicator equal to one for bonds issued in the issuer's own currency. Observations are individual bond issues, weighted by amount issued. Standard errors clustered by country are in parentheses. $^{***}p < 0.01$.

Table 12: Maturity and Currency of Issuance

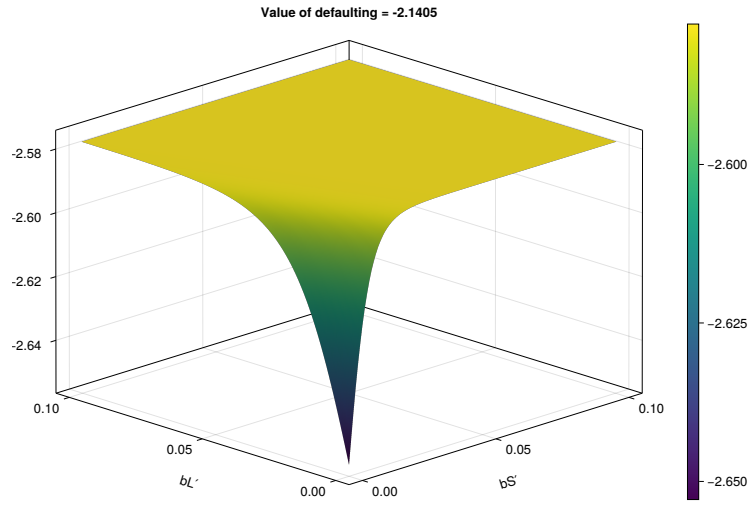


Figure 14: Objective Function Across Different Portfolios in a Default State

C Solution Method

C.A Interpolation Method

I solve the model with value function iteration, where I approximate all equilibrium functions as Gaussian processes. I use Gaussian process regression for three main reasons. First, the number of points required to accurately approximate the equilibrium functions is orders of magnitude lower than even powerful sparse grid methods such as Smolyak's method. Second, Gaussian Process Regression inherently provides uncertainty quantification, which facilitates the use of Bayesian Active Learning to adaptively add training points in regions of the state space where the Gaussian process is most uncertain. In highly nonlinear models such as this one, this adaptivity is extremely useful.

Lastly, and most importantly, Gaussian Process Regression is grid-less. This is particularly crucial when approximating the policy functions for bond issuance because the domain over which the policy functions are well-defined is irregular instead of a standard hypercube.

Figure 14 shows the objective function in the maturity choice model of Arellano and Ramanarayanan (2012) evaluated across a fine grid of potential short-term and long-term bond portfolio choices at a very low wealth (low output and high indebtedness) state. Here, the sovereign would choose to default in equilibrium since the value from defaulting (-2.14) is greater than the maximized value from repaying (-2.58).

Although the maximum value of the objective function is unique, the short-term/long-term portfolio that attains this maximum is not. In such a state, multiple portfolio combinations yield the same revenue raised, and different portfolios will not affect the sovereign's continuation value since it is likely to default in the next period anyway. Thus, the outputted optimal policy at this

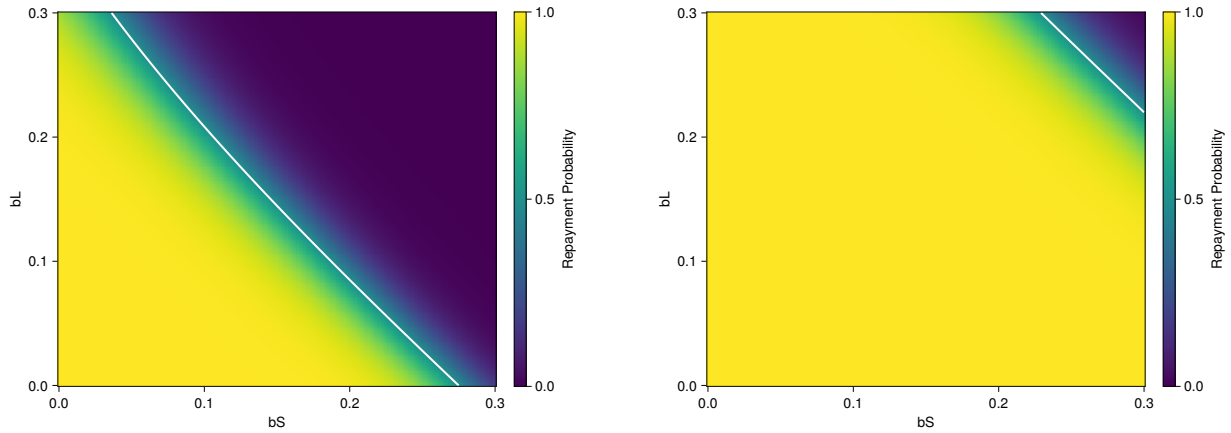


Figure 15: Repayment Heatmap Across Different Output States

grid point would only be numerical noise, and even the tiniest change in the state space often results in the optimizer choosing a completely different portfolio.

Figure 15 plots the probability of repayment across different indebtedness levels for two different output levels, and shows that the set of points that correspond to repayment more closely resembles a simplex than a hypercube, which is what traditional grid-based methods require. Smolyak's method particularly struggles with this issue because it interpolates between points with Chebyshev polynomials, which are global, meaning that the policies evaluated at the default points may contaminate the approximation of the policy functions even at points which correspond to repayment.

The grid-less nature of Gaussian Process Regression is particularly useful in this context because it allows me to only use points that correspond to repayment in equilibrium when training the Gaussian process for the policy functions. This ensures that the policy functions are only trained on meaningful data where the policy functions are well-defined. Meanwhile, standard grid-based methods require the optimization to succeed at all points over the hypercube to construct the interpolant.

C.B Algorithm

As is now standard in the sovereign debt literature, I follow Hatchondo et al. (2010) and set the initial guess to correspond to the terminal period of a finite horizon. However, I further impose that the sovereign buys back all of its unmatured long-term debt in the terminal period. This additional buy-back assumption is important because without it, the maturity of the long-term bonds would initially depend on the number of periods until the terminal horizon. Short-term and long-term debt would be identical in period $T-1$, or the first iteration, which often results in numerical instability.

The buy-backs are done at the price that is consistent with the sovereign defaulting with owing

no debt in period $T+1$, resulting in the period $T+1$ being set to the target level, and there being no dilution risk since the sovereign cannot issue new debt. For period $T+1$ default risk, the value from defaulting is equivalent to the value from repaying for all income levels such that there are no default costs from defaulting, or

$$\mathcal{Y}_{T+1} = h(\mathcal{Y}_{T+1}). \tag{27}$$

For all $\mathcal{Y}_{T+1}$ below this threshold level, the sovereign is indifferent between defaulting and repaying, so under the taste shock assumption, the sovereign's probability of repayment is $\frac{1}{2}$. For all $\mathcal{Y}_{T+1}$ above this threshold level, defaulting will result in a lower value than repaying with no debts owed. Without the taste shock assumption, the sovereign would always choose to repay in such states. With the addition of taste shocks, the discrete jump from $\frac{1}{2}$ to 1 is smoothed out, with the smoothness increasing with the variance of the taste shock.

In subsequent iterations, I first approximate the period $T-1$ pricing functions with the period T functions. Then, I use the period $T-1$ pricing functions and period T value functions to update the period $T-1$ value functions and policy functions. I repeat this process until the value function for repayment converges.

C.C Training Points

The Gaussian processes for all pricing functions, value functions, and the policy function for inflation are trained on a set of points generated from two sources. First, I create a set of points that correspond to an anisotropic Smolyak sparse grid. This set of points stays fixed throughout iterations.

Figure 16 shows the distribution of points generated from a level $\mu = 3$ isotropic Smolyak sparse grid corresponding to the model in Arellano (2008). Smolyak's method tends to place points at the edges of the domain, which is particularly useful for alleviating Gaussian Process Regression's tendency to extrapolate poorly.[16]

The second source of points is generated via Bayesian Active Learning, where I add points in regions of the state space where the Gaussian process for the pricing function for the long-term domestic currency bond is most uncertain. I choose to add points based on the uncertainty of the long-term domestic currency bond pricing function because it is the most nonlinear and difficult to approximate of all four pricing functions.

For the policy functions for bond issuance, I only use points that correspond to repayment in equilibrium for the aforementioned reason. After solving the Bellman equation at the aforementioned points, I evaluate the value functions for repayment and default at each of them and only include points where the value from repaying is higher than the value from defaulting.

[16] The nodes are allocated according to the Clenshaw-Curtis grid structure. The alternative, Chebyshev–Gauss–Lobatto, tends to place *too* many points at the edges of the hypercube, which is often inefficient.

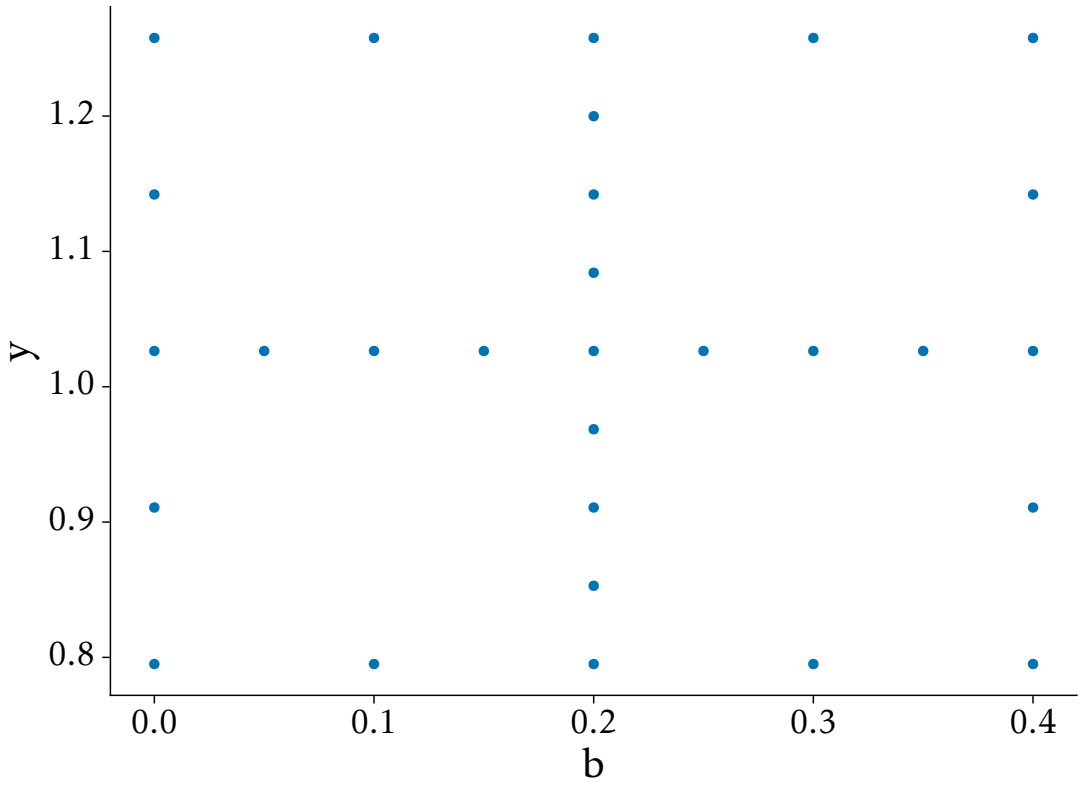


Figure 16: Smolyak Sparse Grid Points

C.D Gaussian Process Details

I use the Matérn 3/2 kernel for all Gaussian processes with automatic relevance determination and set the noise to 10^{-6} for all Gaussian processes, rather than including it as a hyperparameter to be optimized. I further follow Gu and Stangebye (2023) by linearly scaling all states to be in the unit hypercube, and applying a logit transformation on all bounded functions before applying the Gaussian Process Regression.